

COMPLETE STUDY HIERARCHY · TRACKS A-E

Master Software Engineering Curriculum

A book-style syllabus for DSA, frontend engineering, Java and backend engineering, low-level and system design, distributed systems, and applied AI engineering.

5	591	1281	4661
TRACKS	SECTIONS	CORE TOPICS	TOTAL STUDY NODES

Generated September 4, 2026 from the live website taxonomy. Core topics are classified curriculum items; total study nodes additionally include nested concepts. NeetCode and CSES problem banks are represented by their curriculum categories rather than reproducing every individual problem.

Curriculum Summary

Read each track as a part, each named group as a chapter, each numbered section as a syllabus unit, and the indented entries as subtopics.

A	DSA & Problem Solving	14 sections · 117 core · 135 nodes
B	Frontend Engineering	165 sections · 647 core · 1658 nodes
C	Java & Backend Engineering	255 sections · 329 core · 1941 nodes
D	LLD, System Design & Distributed Systems	80 sections · 111 core · 456 nodes
E	Applied AI Engineering	77 sections · 77 core · 471 nodes

DSA & Problem Solving

Algorithms, patterns, and interview coding execution. 14 sections · 135 study nodes.

A1 — Programming for Interviews

16 nodes

Java Syntax for Competitive Coding	TIER 1
Collections Framework	TIER 1
— HashMap	
— HashSet	
— TreeMap	
— TreeSet	
— Deque	
— PriorityQueue	
Comparable / Comparator	TIER 1
Strings / StringBuilder	TIER 1
Custom Classes	TIER 1
Recursion	TIER 1
Time/Space Complexity	TIER 1
— Space Complexity	
Big-O Analysis	TIER 1
Java Streams for Interviews	TIER 2

A2 — Arrays & Strings

11 nodes

Array/String Traversal and Manipulation	TIER 1
Prefix/Suffix Techniques	TIER 1
Two Pointers	TIER 1
Sliding Window	TIER 1
Frequency Counting	TIER 1
Hashing	TIER 1
Sorting-Based Problems	TIER 1
Matrix Problems	TIER 1
Interval Problems	TIER 1
Kadane's Algorithm	TIER 1
Difference Arrays	TIER 2

A3 — Searching & Sorting

9 nodes

Binary Search	TIER 1
Binary Search on Answer	TIER 1
Lower/Upper Bounds	TIER 1
Merge Sort	TIER 1
Quick Sort Concepts	TIER 1
Comparator-Based Sorting	TIER 1
Quickselect	TIER 2
Counting/Bucket Sorting Concepts	TIER 2

A4 — Linked Lists

8 nodes

Singly/Doubly Linked Lists	TIER 1
— Doubly Linked Lists	
Fast/Slow Pointers	TIER 1
Linked List Reversal	TIER 1
Cycle Detection	TIER 1
Merge Lists	TIER 1
List Intersection	TIER 1
LRU-Style Linked-List Manipulation	TIER 1

A5 — Stack & Queue

8 nodes

Stack	TIER 1
Queue	TIER 1
Deque	TIER 1
Monotonic Stack	TIER 1
Monotonic Queue	TIER 1
Next Greater/Smaller Element	TIER 1
Parentheses/Expression Problems	TIER 1
— Expression Problems	

A6 — Trees

13 nodes

Binary Trees	TIER 1
Tree DFS	TIER 1
Tree BFS/Level-Order Traversal	TIER 1
— Level-Order Traversal	
Binary Search Trees	TIER 1
Tree Height/Depth	TIER 1
Tree Diameter	TIER 1
Lowest Common Ancestor	TIER 1
Tree Path Problems	TIER 1
Tree Construction	TIER 1
Serialization/Deserialization	TIER 1
Tree Views	TIER 2
N-ary Trees	TIER 2

A7 — Heaps

7 nodes

Min/Max Heap	TIER 1
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— Max Heap	
Priority Queues	TIER 1
Top-K Problems	TIER 1
K-way Merge	TIER 1
Running Median	TIER 1
Two-Heap Pattern	TIER 2

A8 — Graphs

15 nodes

Graph Representation	TIER 1
Graph BFS/DFS	TIER 1
— Graph DFS	
Connected Components	TIER 1
Graph Cycle Detection	TIER 1
Topological Sorting	TIER 1
Multi-source BFS	TIER 1
Union-Find / DSU	TIER 1
Dijkstra	TIER 1
Bellman-Ford	TIER 2
Floyd-Warshall	TIER 2
Minimum Spanning Tree (Kruskal/Prim)	TIER 2
— MST — Prim	
Bipartite Graphs	TIER 2
Strongly Connected Components	TIER 2

A9 — Recursion & Backtracking

8 nodes

Subsets	TIER 1
Permutations	TIER 1
Combinations	TIER 1
Combination Sum	TIER 1
Grid Search Backtracking	TIER 1
N-Queens	TIER 1
Constraint-Based Backtracking	TIER 1
Pruning	TIER 1

A10 — Greedy

4 nodes

Interval Scheduling	TIER 1
Sorting + Greedy	TIER 1
Heap + Greedy	TIER 1
Local vs Global Optimum Reasoning	TIER 1

A11 — Dynamic Programming

13 nodes

Memoization vs Tabulation	TIER 1
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— Tabulation

1D DP	TIER 1
2D DP	TIER 1
Knapsack Patterns	TIER 1
Subsequence Problems	TIER 1
Longest Increasing Subsequence	TIER 1
Longest Common Subsequence	TIER 1
Grid DP	TIER 1
Partition DP Basics	TIER 1
DP on Trees	TIER 2
State-Machine DP	TIER 2
Bitmask DP	TIER 3

A12 — Tries & Advanced Structures

5 nodes

Trie	TIER 2
Prefix Matching	TIER 2
Segment Trees	TIER 2
Fenwick Tree	TIER 3
Sparse Tables	TIER 3

A13 — Bit Manipulation

7 nodes

AND / OR / XOR	TIER 1
Set/Unset/Check Bit	TIER 1
— Unset Bit	
— Check Bit	
Powers of Two	TIER 1
XOR Tricks	TIER 2
Bitmasking	TIER 2

A14 — Coding Interview Execution

11 nodes

Clarifying Requirements	TIER 1
Explaining Brute-Force First	TIER 1
Deriving Optimized Solution	TIER 1
Thinking Aloud	TIER 1
Clean Coding Without IDE Assistance	TIER 1
Manual Testing	TIER 1
Edge-Case Identification	TIER 1
Complexity Analysis in Interviews	TIER 1
Responding to Interviewer Hints	TIER 1
Timed Mock Interviews	TIER 1
— Mock Interviews	

Frontend Engineering

JavaScript language (beginner → interview), TypeScript, React, browser APIs. 165 sections · 1658 study nodes.

B1 — JavaScript

43 sections

B1.1 — JavaScript Foundations

23 nodes

What JavaScript Is	TIER 1
JavaScript vs ECMAScript	TIER 1
JavaScript Runtimes	TIER 1
— Browser Runtime	
— JavaScript Engine	
— Web APIs vs Language	
— Node.js Conceptually	
How JavaScript Loads in a Webpage	TIER 1
— Inline Scripts	
— External Scripts	
— script async	
— script defer	
— type="module"	
Statements vs Expressions	TIER 1
Syntax Basics	TIER 1
— Comments	
— Identifiers	
— Keywords and Reserved Words	
— Case Sensitivity	
— Semicolons / ASI	
— Strict Mode	
— Console and Output	
Basic Debugging	TIER 1

B1.2 — Variables & Declarations

15 nodes

Variables	TIER 1
— Declaration	
— Initialization	
— Assignment vs Reassignment	
— Redefinition	
var	TIER 1
let	TIER 1
const	TIER 1
var vs let vs const	TIER 1
— Function Scope	
— Block Scope	
— Global Scope	
Temporal Dead Zone	TIER 1
Global Variables and globalThis	TIER 1
Variable Naming / Best Practices	TIER 2

B1.3 — Types & Values

23 nodes

Primitive Types	TIER 1
— string	

— number	
— bigint	
— boolean	
— undefined	
— null	
— symbol	
Objects as Values	TIER 1
Primitive vs Reference Types	TIER 1
— Primitive Immutability	
— Object Mutability	
— Call-by-Sharing / Parameter Passing	
typeof	TIER 1
— typeof null	
undefined vs null vs Undeclared	TIER 1
NaN, Infinity, and Safe Numbers	TIER 1
— Number.isNaN()	
— Object.is()	
Wrapper Objects / Autoboxing	TIER 1
Truthy and Falsy Values	TIER 1
instanceof	TIER 2
Symbol.toStringTag	TIER 2

B1.4 — Type Conversion & Coercion 17 nodes

Explicit Conversion	TIER 1
— String()	
— Number()	
— Boolean()	
— parseInt / parseFloat	
— Unary +	
Type Coercion	TIER 1
— Implicit Coercion	
— Coercion with +	
Equality Operators	TIER 1
— == Abstract Equality	
— === Strict Equality	
— != and !==	
Object-to-Primitive Conversion	TIER 2
— valueOf()	
— toString()	
— Symbol.toPrimitive	

B1.5 — Operators & Expressions 14 nodes

Arithmetic Operators	TIER 1
Assignment Operators	TIER 1
Comparison Operators	TIER 1
Logical Operators	TIER 1
— Short-Circuit Evaluation	
Unary Operators	TIER 1
— Increment / Decrement	
Ternary Operator	TIER 1
Operator Precedence & Associativity	TIER 1
Optional Chaining	TIER 1
Nullish Coalescing	TIER 1

— Logical Assignment (&&= = ??=)	TIER 2
Bitwise Operators	TIER 2
in, delete, void, instanceof	TIER 2
B1.6 — Control Flow	7 nodes
if / else / else if	TIER 1
switch	TIER 1
for / while / do...while	TIER 1
for...of vs for...in	TIER 1
break / continue	TIER 1
— Labels	
Nested Control Flow	TIER 1
B1.7 — Strings	12 nodes
String Creation & Immutability	TIER 1
— Indexing and Length	
Core String Methods	TIER 1
— Searching Strings	
— Extracting / Slicing	
— Replacing / Splitting / Trimming	
Template Literals	TIER 1
— Interpolation	
— Multiline Strings	
— Escape Sequences	
Tagged Template Literals	TIER 2
Unicode / UTF-16 Basics	TIER 2
B1.8 — Numbers & Math	8 nodes
Number Representation	TIER 1
— Floating-Point and 0.1 + 0.2	
— Safe Integers	
Number Methods & Parsing	TIER 1
Math	TIER 1
— Rounding	
— Random Numbers	
BigInt	TIER 2
B1.9 — Functions	17 nodes
Function Declaration	TIER 1
Function Expression	TIER 1
— Named Function Expression	
— Anonymous Function	
Arrow Functions	TIER 1
— Lexical this	
Parameters, Defaults, Rest, arguments	TIER 1
First-Class / Higher-Order Functions	TIER 1
— Callbacks	
Pure vs Impure Functions	TIER 1
— Side Effects	
IIFE	TIER 1
Function Hoisting vs Expression	TIER 1
Recursion	TIER 1
Currying & Partial Application	TIER 2

— compose / pipe	
Memoization	TIER 2
B1.10 — Scope & Lexical Environments	7 nodes
Scope	TIER 1
— Module Scope	
Lexical Scope	TIER 1
— Scope Chain	
— Lexical Environment	
— Shadowing	
Variable Resolution	TIER 1
B1.11 — Hoisting & Temporal Dead Zone	6 nodes
Hoisting	TIER 1
— Creation vs Execution Phase	
— var Hoisting	
— let/const Hoisting	
— Class Hoisting	
Hoisting Interview Puzzles	TIER 1
B1.12 — Closures	9 nodes
Closures	TIER 1
— Closure Lifecycle	
— Private State / Function Factories	
Loops + Closures	TIER 1
— var Loop Problem	
Stale Closures	TIER 1
— Closures in React	
Closure Memory Implications	TIER 1
Closure Interview Puzzles	TIER 1
B1.13 — Objects	12 nodes
Object Literals	TIER 1
— Dot vs Bracket Notation	
— Computed Property Names	
— Property / Method Shorthand	
Object APIs	TIER 1
— keys / values / entries / fromEntries	
— Object.assign / Object.create	
— freeze / seal / preventExtensions	
— hasOwn / ownership	
Shallow vs Deep Copy	TIER 1
— structuredClone	
— Why JSON Cloning Is Problematic	
B1.14 — Object Property Model	6 nodes
Own vs Inherited Properties	TIER 1
— Enumerable Properties	
Property Descriptors	TIER 2
— Object.defineProperty	

— writable / enumerable / configurable	TIER 2
Getters & Setters	
B1.15 — this, call, apply, bind	11 nodes
this — Default / Global Binding — Implicit Binding — Explicit Binding — new Binding — Arrow-Function Lexical Binding — Losing this in Callbacks	TIER 1
call, apply, bind — apply — bind — Function Borrowing	TIER 1
B1.16 — Prototypes & Prototype Chain	9 nodes
Prototypes — [[Prototype]] / getPrototypeOf	TIER 1
Prototype Chain — Property Lookup & Shadowing	TIER 1
Constructor Functions — What new Actually Does — constructor Property	TIER 1
Object.create	TIER 1
Prototype Pollution (Concept)	TIER 1
B1.17 — Classes & OOP	8 nodes
Classes — constructor — Instance vs Static Methods — Public / Private Fields	TIER 1
extends / super / Inheritance — Method Overriding	TIER 1
Encapsulation, Polymorphism, Composition	TIER 1
Classes vs Factories vs Constructors	TIER 1
B1.18 — Arrays	14 nodes
Array Creation, Indexing, Length — Sparse Arrays	TIER 1
Mutating vs Non-Mutating Methods — push / pop / shift / unshift — slice vs splice	TIER 1
Searching Arrays — includes / indexOf / find — some / every	TIER 1
map / filter / reduce — forEach	TIER 1

— flat / flatMap	
sort and JavaScript Sorting Gotchas	TIER 1
Array.from / Array.isArray	TIER 1
Copying Arrays	TIER 2
B1.19 — Maps, Sets & Weak Collections	7 nodes
Set	TIER 1
— Set vs Array	
Map	TIER 1
— Map vs Object	
WeakMap	TIER 2
— Weak References & GC	
WeakSet	TIER 2
B1.20 — Destructuring, Spread & Modern Syntax	3 nodes
Destructuring	TIER 1
— Defaults / Renaming / Nested	
Spread/Rest	TIER 1
B1.21 — Iterables, Iterators & Generators	6 nodes
Iterators/Generators	TIER 2
— Iterable / Iterator Protocol	
— Symbol.iterator	
— Generators	
— yield / yield*	
— Async Iterators / for await...of	
B1.22 — Symbols	3 nodes
Symbols	TIER 2
— Symbol.for / keyFor	
— Well-known Symbols	
B1.23 — Execution Context	4 nodes
Execution Contexts	TIER 1
— Global Execution Context	
— Function Execution Context	
— Variable / Lexical Environment	
B1.24 — Call Stack	4 nodes
Call Stack	TIER 1
— Stack Frames	
— Stack Overflow	
— Reading Stack Traces	
B1.25 — JavaScript Runtime & Engine	6 nodes
JavaScript Engine	TIER 1
— Engine vs Host Environment	
— Heap	
Parsing, AST, Interpreter, JIT	TIER 2
— V8 Architecture Conceptually	

B1.26 — Event Loop	7 nodes
JavaScript Event Loop	TIER 1
— Run-to-Completion	
Microtasks vs Macrotasks	TIER 1
— Macrotasks	
— queueMicrotask	
— Microtask Starvation	
Event-Loop Interview Puzzles	TIER 1
B1.27 — Timers & Scheduling	4 nodes
setTimeout / clearTimeout	TIER 1
— Minimum-Delay Misconception	
setInterval / clearInterval	TIER 1
requestAnimationFrame (Language View)	TIER 2
B1.28 — Callbacks	4 nodes
Callback Concept	TIER 1
— Sync vs Async Callbacks	
— Error-First Callback Pattern	
Callback Hell & Inversion of Control	TIER 1
B1.29 — Promises	11 nodes
Promises	TIER 1
— Pending / Fulfilled / Rejected	
— .then / .catch / .finally	
— Promise Chaining	
Error Propagation	TIER 1
Promise Combinators	TIER 1
— Promise.all	
— Promise.allSettled	
— Promise.race	
— Promise.any	
Sequential vs Parallel Async Work	TIER 1
B1.30 — Async/Await	4 nodes
async/await	TIER 1
— try/catch with await	
— Why await Does Not Block the Runtime	
Async Control Flow	TIER 1
B1.31 — Async Cancellation & Coordination	4 nodes
AbortController / AbortSignal (Language)	TIER 2
Race Conditions & Stale Requests	TIER 2
— Timeouts / Retries / Deduplication	
Concurrency Limiting	TIER 2
B1.32 — Error Handling	5 nodes
Error Types	TIER 1

— TypeError / ReferenceError / RangeError	
throw / try / catch / finally	TIER 1
Custom Errors	TIER 1
Unhandled Promise Rejection	TIER 1
B1.33 — Debugging	6 nodes
Browser DevTools	TIER 1
— Breakpoints & Stepping	
— Watch / Scope / Call Stack	
Async Debugging	TIER 1
Source Maps Conceptually	TIER 1
Performance & Memory Profiling Basics	TIER 1
B1.34 — Modules	10 nodes
Modules	TIER 1
— export / import	
— Default vs Named Exports	
— Dynamic import()	
ESM vs CommonJS	TIER 1
— Live Bindings	
— Circular Dependencies	
— Top-level await	
— Tree Shaking Conceptually	
IIFE / Module Pattern (Historical)	TIER 2
B1.35 — JSON & Serialization	4 nodes
JSON vs JavaScript Objects	TIER 1
— JSON.stringify / parse	
— Unsupported Values & Circular Refs	
replacer / reviver / toJSON	TIER 2
B1.36 — Dates & Time	6 nodes
Date Basics	TIER 1
— Unix Timestamps	
— Local Time vs UTC	
Date Arithmetic Pitfalls	TIER 2
Intl.DateTimeFormat	TIER 2
Temporal API Direction	TIER 3
B1.37 — Regular Expressions	4 nodes
Regex Literals & Constructor	TIER 2
— Classes, Quantifiers, Anchors, Flags	
test / exec / match / replace	TIER 2
Groups & Lookaround Basics	TIER 2
B1.38 — Functional JavaScript	3 nodes
Declarative vs Imperative Style	TIER 1
— Immutability	
Referential Transparency Concept	TIER 2
B1.39 — Memory Management & Garbage Collection	6 nodes
Memory Management	TIER 1

— Reachability & Roots	
Garbage Collection	TIER 1
— Mark-and-Sweep / Generational GC	
Common Memory Leaks	TIER 1
— Heap Snapshots	

B1.40 — Performance Patterns 4 nodes

Debouncing	TIER 1
— Debounce vs Throttle	
Throttling	TIER 1
Long Tasks / Main-Thread Blocking	TIER 2

B1.41 — Proxy & Reflect 4 nodes

Proxy/Reflect	TIER 2
— Reflect	
— Traps (get/set/has/deleteProperty)	
— Validation / Reactivity Concepts	

B1.42 — Binary Data 2 nodes

ArrayBuffer / TypedArray / DataView	TIER 3
Blob / File / FileReader (Language)	TIER 3

B1.43 — JavaScript Implementation Exercises 11 nodes

Implement map / filter / reduce / forEach	TIER 1
Implement call / apply / bind	TIER 1
Implement Promise.all / allSettled / race	TIER 1
Implement debounce / throttle / memoize	TIER 1
Implement curry / compose / pipe	TIER 1
Implement deep clone / flatten / deep equality	TIER 1
Implement get(object, path)	TIER 1
Implement EventEmitter	TIER 1
Implement retry / sleep / concurrency limiter	TIER 1
Implement a Promise	TIER 2
Implement LRU Cache in JavaScript	TIER 2

B2 — TypeScript 35 sections

B2.1 — Introduction & Setup 7 nodes

What TypeScript Is	TIER 1
TypeScript vs JavaScript	TIER 1
Why Use TypeScript	TIER 1
TypeScript Trade-offs	TIER 2
Optional Static Typing	TIER 1
Compiling with tsc	TIER 1
.ts vs .tsx	TIER 1

B2.2 — Simple Types & Annotations 6 nodes

Primitive & Object Types	TIER 1
— string	
— number	

— boolean	
Explicit Types	TIER 1
Type Annotations	TIER 1
B2.3 — Type Inference	2 nodes
Type Inference	TIER 1
When to Annotate vs Infer	TIER 1
B2.4 — Special Types	7 nodes
any, unknown, never, void	TIER 1
— Avoiding any	
— unknown	
— never	
— void	
null & undefined	TIER 1
strictNullChecks	TIER 1
B2.5 — Arrays	3 nodes
Typed Arrays	TIER 1
— Readonly Arrays	
— Array Type Inference	
B2.6 — Tuples	2 nodes
Tuples	TIER 1
Readonly Tuples	TIER 1
B2.7 — Object Types	4 nodes
Object Types	TIER 1
— Optional Properties	
— readonly Properties	
Parameter Destructuring	TIER 1
B2.8 — Enums & Literal Constants	4 nodes
Enums	TIER 1
— String Enums	
Literal Unions vs Enums	TIER 1
as const Objects	TIER 1
B2.9 — Interfaces	3 nodes
Interfaces	TIER 1
— Extending Interfaces	
Interface vs Type — When to Use	TIER 1
B2.10 — Type Aliases	2 nodes
Type Aliases	TIER 1
Type Alias Patterns	TIER 1
B2.11 — Unions & Intersections	3 nodes
Union Types	TIER 1

— Intersection Types	
Union & Intersection Patterns	TIER 1
B2.12 — Literal Types & Discriminated Unions	4 nodes
Literal Types	TIER 1
Discriminated Unions	TIER 1
— Exhaustiveness & assertNever	
RemoteData Async State Pattern	TIER 1
B2.13 — Type Narrowing & Type Guards	5 nodes
Type Narrowing	TIER 1
— typeof Narrowing	
— instanceof Narrowing	
— in Operator	
— Custom Type Guards	
B2.14 — Functions	5 nodes
Typed Functions	TIER 1
— Optional & Rest Parameters	
— Arrow Functions	
Function Overloading	TIER 1
Function Type Variance	TIER 2
B2.15 — Casting & Assertions	3 nodes
Type Assertions (as)	TIER 1
— as const Assertion	
— Non-null Assertion (!)	
B2.16 — Classes	3 nodes
Classes	TIER 1
— Class Inheritance	
— super & Constructor	
B2.17 — Access Modifiers & Encapsulation	3 nodes
public / private / protected	TIER 1
— readonly Members	
Parameter Properties	TIER 1
B2.18 — Abstract Classes & implements	4 nodes
Abstract Classes	TIER 1
— Method Overriding	
implements vs extends	TIER 1
Singleton Pattern	TIER 2
B2.19 — Generics	4 nodes
Generics	TIER 1
— Generic Classes	

— Default Type Parameters	
Generic Constraints	TIER 1
B2.20 — Utility Types	10 nodes
Utility Types Overview	TIER 1
— Partial	
— Required	
— Pick	
— Omit	
— Record	
— Readonly	
— ReturnType	
— Parameters	
— Awaited	
B2.21 — keyof, typeof & Indexed Access	4 nodes
keyof	TIER 1
typeof (type level)	TIER 1
Indexed Access Types	TIER 1
keyof + typeof Patterns	TIER 1
B2.22 — Structural Typing & Declaration Merging	3 nodes
Structural Typing	TIER 1
Declaration Merging	TIER 2
— Module Augmentation	
B2.23 — Index Signatures	2 nodes
Index Signatures	TIER 1
Record vs Index Signature	TIER 1
B2.24 — satisfies & as const	3 nodes
satisfies	TIER 1
as const	TIER 1
Variant Maps with satisfies	TIER 1
B2.25 — Conditional Types & infer	3 nodes
Conditional Types	TIER 2
infer	TIER 2
Async Return with infer / Awaited	TIER 2
B2.26 — Mapped & Template Literal Types	4 nodes
Mapped Types	TIER 2
— Key Remapping (as)	
Template Literal Types	TIER 2
ChangeHandlers Pattern	TIER 2
B2.27 — Type-Safe API Design	3 nodes
Type-Safe API Design	TIER 1

— Pick + Partial Patch Payloads	
Generic pick() Helper	TIER 1
B2.28 — React + TypeScript	5 nodes
React + TypeScript	TIER 1
— Component Props	
— Controlled vs Uncontrolled	
Mutually Exclusive Props	TIER 1
Typing Hooks	TIER 1
B2.29 — Node.js + TypeScript	2 nodes
Node.js + TypeScript	TIER 1
— Express Request/Response Typing	
B2.30 — Modules	3 nodes
TypeScript Modules	TIER 1
ESM vs CommonJS in TS	TIER 1
Namespaces	TIER 2
B2.31 — tsconfig & Compiler Options	5 nodes
tsconfig.json	TIER 1
— strict & Compiler Flags	
— noImplicitAny	
Declaration Files (.d.ts)	TIER 1
Source Maps & Debugging	TIER 1
B2.32 — Tooling & Ecosystem	4 nodes
Definitely Typed (@types)	TIER 1
Third-Party Library Types	TIER 1
ESLint + TypeScript	TIER 1
Vite / Build Tooling	TIER 1
B2.33 — Migration & Decorators	3 nodes
Migrating JavaScript → TypeScript	TIER 1
— allowJs & checkJs	
Decorators	TIER 2
B2.34 — Advanced Patterns	3 nodes
Mixins	TIER 2
this Types	TIER 2
Branded / Nominal Types	TIER 2
B2.35 — TypeScript Implementation Exercises	5 nodes
Implement type-safe pick()	TIER 1
Implement RemoteData<T>	TIER 1
Implement Exclusive Button Props	TIER 1
Implement ChangeHandlers<T>	TIER 2
Implement isPreferences Guard	TIER 1

B3.1 — Browser Architecture

24 nodes

Browser Architecture Overview	TIER 1
Browser Processes	TIER 1
— Browser Process	
— Renderer Process	
— GPU Process	
— Network Process	
— Utility / Plugin Processes	
Browser Threads	TIER 1
— Main Thread	
— Compositor Thread	
— Raster Threads	
— Worker Threads in the Browser	
URL Navigation Lifecycle	TIER 1
— DNS Resolution	
— TCP & TLS Handshake	
— HTTP Request & Response	
— Document Download & Parsing Start	
Site Isolation & Process Model	TIER 1
— Origin-Keyed Processes	
— Cross-Site iframe Process Boundaries	
Rendering Engine (Blink / WebKit)	TIER 1
JavaScript Engine in the Browser	TIER 1
Browser Security Sandbox	TIER 1
Multi-Tab & Multi-Window Architecture	TIER 2

B3.2 — Window, Document & BOM

29 nodes

Browser Object Model (BOM)	TIER 1
window Object	TIER 1
— Global Scope & window	
— Window Properties & Methods	
— window.open / close / focus	
document Object	TIER 1
— document Properties	
— document Methods	
— document.readyState	
globalThis in the Browser	TIER 1
navigator Object	TIER 1
— userAgent & Client Hints	
— language & languages	
— onLine & Connection Info	
— clipboard (Overview)	
location Object	TIER 1
— href / origin / pathname / search / hash	
— assign / replace / reload	
history Object	TIER 1
— History Stack Model	
— back / forward / go	
History API	TIER 1
— pushState / replaceState	
— popstate Event	

— hashchange Event	
screen Object	TIER 2
console in the Browser	TIER 1
setTimeout / setInterval in Browser	TIER 1
alert / prompt / confirm	TIER 2

B3.3 — DOM Fundamentals 34 nodes

DOM	TIER 1
— DOM Elements	
DOM Tree Model	TIER 1
— Node Types (Element, Text, Comment, Document)	
— Parent / Child / Sibling Relationships	
— DocumentFragment	
Selecting Elements	TIER 1
— querySelector / querySelectorAll	
— getElementById	
— getElementsByClassName	
— getElementsByTagName	
— matches / closest	
DOM Manipulation	TIER 1
— createElement / createTextNode	
— append / prepend / appendChild	
— insertBefore / insertAdjacentElement	
— remove / replaceChild / replaceWith	
— cloneNode	
DOM Content & Text	TIER 1
— textContent vs innerText	
— innerHTML / outerHTML	
— insertAdjacentHTML	
Attributes & Properties	TIER 1
— getAttribute / setAttribute / removeAttribute	
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— renderHook — act	
Snapshot Testing & Limitations	TIER 2
E2E with Playwright / Cypress	TIER 2

B4.21 — Accessibility in React 10 nodes

React Accessibility	TIER 1
— Semantic Components — Accessible Forms & Labels	

- Keyboard Interactions
- ARIA in React
- Accessible Modals
- Accessible Custom Widgets
- used for Accessibility
- Route-Change Accessibility
- Accessibility Testing Concepts

B4.22 — Internationalization

10 nodes

Internationalization in React

TIER 2

- i18n vs l10n
- Translation Architecture
- Locale Management
- Intl with React
- react-i18next / react-intl Concepts
- Numbers, Dates & Currency
- Pluralization
- RTL Layouts
- Lazy-Loading Translations

B4.23 — Legacy React & Existing Codebases

14 nodes

Reading Class Components

TIER 2

- this.props
- this.state & setState
- constructor

Class Lifecycle Methods

TIER 2

- componentDidMount
- componentDidUpdate
- componentWillUnmount
- shouldComponentUpdate
- PureComponent
- Error-Boundary Lifecycle Methods

Legacy & Deprecated APIs

TIER 3

- Legacy Context — Awareness
- Deprecated Lifecycle Methods

B4.24 — React Architecture & Production Engineering

25 nodes

Component Boundaries

TIER 1

- Feature-Based Organization
- Container / UI Separation
- Custom-Hook Architecture
- State Ownership
- Dependency Direction

Production UI Architecture

TIER 1

- API & Data Layer
- Error Architecture
- Loading-State Architecture
- Authentication & Authorization Integration
- Feature Flags
- Configuration & Environments
- Design-System Integration

Large-Scale React Application Structure

TIER 1

- Performance Budgets
- Observability & Error Monitoring
- Analytics Integration
- Maintainability & Refactoring

React Anti-Patterns

TIER 1

- Direct State Mutation
- Effect Misuse

- Bad Keys
- Conditional Hooks
- Excessive Global State

B5 — CSS / UI Engineering

12 sections

B5.1 — Cascade & Selectors

9 nodes

The Cascade

- Origin, Importance & Source Order
- Inheritance
- !important

Specificity

- Specificity Calculation
- :is() & :where()
- :has()

@layer Cascade Layers

TIER 1

TIER 1

TIER 2

B5.2 — Box Model & Flow

7 nodes

Box Model

- box-sizing
- Margin Collapse
- Overflow
- min-width: 0 & Flex/Grid Shrink Bugs

Block Formatting Context

- Containing Floats

TIER 1

TIER 1

B5.3 — Display & Visibility

4 nodes

The display Property

- block vs inline
- inline-block
- none vs visibility vs opacity

TIER 1

B5.4 — Flexbox

6 nodes

Flexbox

- Main Axis vs Cross Axis
- Wrapping
- flex-grow, flex-shrink & flex-basis
- Alignment & Centering
- When Flexbox vs Grid

TIER 1

B5.5 — Grid

6 nodes

Grid

- Tracks, fr & gap
- minmax()
- Grid Template Areas
- auto-fit vs auto-fill
- Subgrid

TIER 1

B5.6 — Positioning & Stacking

8 nodes

Positioning

- Containing Block
- relative & absolute
- position: fixed

TIER 1

— position: sticky	
Stacking Context	TIER 1
— z-index	
— What Creates a Stacking Context	
B5.7 — Units, Color & Typography	9 nodes
CSS Units	TIER 1
— px vs rem vs em	
— Viewport Units (vw, svh)	
— clamp() & Fluid Values	
Typography	TIER 1
— Line Height & Measure	
— Web Fonts	
Color	TIER 1
— color-scheme & light-dark()	
B5.8 — Responsive Design	6 nodes
Responsive Design	TIER 1
— Mobile-First	
— Media Queries	
— Fluid Type & Spacing	
Container Queries	TIER 1
Logical Properties	TIER 1
B5.9 — Theming & Custom Properties	3 nodes
Custom Properties	TIER 1
— Theming with CSS Variables	
— Tokens vs Preprocessor Variables	
B5.10 — Motion	5 nodes
Transitions	TIER 1
— Compositor-Safe Animation	
Animations	TIER 1
— @keyframes	
— prefers-reduced-motion	
B5.11 — CSS Architecture	7 nodes
CSS Architecture	TIER 1
— BEM	
— CSS Modules	
— Utility-First / Tailwind	
— CSS-in-JS Trade-offs	
Design Systems	TIER 2
— Design Tokens	
B5.12 — HTML & CSS Accessibility	8 nodes
Semantic HTML	TIER 1
— Landmarks & Document Outline	
— Accessible HTML Forms	
Accessibility	TIER 1
— Focus Styles	
— forced-colors	
ARIA Basics	TIER 1
— When Not to Use ARIA	

B6.1 — Interview Method

13 nodes

RADIO Framework

TIER 1

- Requirements & Constraints
- Architecture Sketch
- Data Model & Ownership
- Interface & API Contracts
- Optimizations (Perf, A11y, i18n)

Question Types: Apps vs Widgets

TIER 1

- Application / Product HLD
- UI Component / Widget LLD
- Hybrid: Widget Inside a Product

Evaluation Criteria & Timeboxing

TIER 1

- Common Interview Mistakes
- Time Split in a 45-Minute Round

B6.2 — Application Architecture

12 nodes

Large-Application Architecture

TIER 1

- SPA vs MPA Trade-offs
- CSR, SSR, SSG & Streaming Choice
- Feature Boundaries & Folder Structure
- Config-Driven UI

Frontend Error Handling

TIER 1

- Error Boundaries at Product Edges
- Retry, Timeout & Fallback UI
- Empty, Partial & Degraded States

SEO-Aware Frontend Architecture

TIER 2

- SSR, Metadata & Crawlability
- Forms, Checkout & Conversion UX

B6.3 — Component & Widget LLD

13 nodes

Component Architecture

TIER 1

- Compound & Headless Components
- Composition vs Configuration
- Shimmer / Skeleton Loading
- Autocomplete / Typeahead
- Dropdown Menu
- Image Carousel
- Modal Dialog
- Data Table
- Poll Widget
- Accordion
- Nested Comments
- Rich Text Editor

B6.4 — State Architecture

5 nodes

State Architecture

TIER 1

- Client, Server & URL State
- Store Boundaries & Ownership
- Optimistic UI & Rollback
- Derived vs Normalized State

B6.5 — API Layer

5 nodes

API-Layer Design

TIER 1

- BFF & Aggregation

- REST vs GraphQL vs RPC at the Client
- List Contracts: Offset, Cursor, Relay
- Error Envelopes & Idempotency

B6.6 — Frontend Authentication

5 nodes

Frontend Authentication

TIER 1

- Session Cookies vs Tokens on the Client
- Protected Routes & Auth Gates
- Authorization in the UI
- Refresh, Logout & Session Expiry

B6.7 — Lists, Search & Pagination

8 nodes

Pagination / Infinite Scrolling

TIER 1

- Infinite Scrolling
- Cursor vs Offset Pagination
- Windowing & Virtualization

Search & Autocomplete Architecture

TIER 1

- Debounce, Cancel & Stale Responses
- Ranking, Empty States & No-Results
- Prefetch & Typeahead Caching

B6.8 — Realtime UI

5 nodes

Real-time UI

TIER 1

- Polling vs WebSocket vs SSE
- Presence, Typing & Read Receipts
- Ordering, Gaps & Replay
- Reconnect, Backoff & Fan-out

B6.9 — Frontend Performance

6 nodes

Frontend Performance

TIER 1

- Core Web Vitals as Design Constraints
- Rendering Patterns at Product Scale
- Network & Build Optimization Choices
- Performance Budgets & Monitoring
- Main-Thread Work & Interaction Latency

B6.10 — Caching & Offline

7 nodes

Frontend Caching

TIER 1

- HTTP Caching Strategy
- API Cache & Stale-While-Revalidate
- In-Memory Cache & Request Deduping

Offline & PWA Strategy

TIER 1

- Service Worker Caching Strategies
- Offline Queue, Sync & Conflict UI

B6.11 — Frontend Security Architecture

5 nodes

Frontend Security Architecture

TIER 1

- XSS & CSRF as Design Choices
- Headers, SRI & iframe Isolation
- Third-Party Scripts & Supply Chain
- Untrusted HTML, Markdown & Embeds

B6.12 — Observability

4 nodes

Frontend Observability

TIER 1

- Telemetry & Real User Monitoring

- Alerting on Client Failures
- Session Replay & Privacy

B6.13 — A11y, i18n & Design Systems

9 nodes

Accessibility in Large Apps

TIER 1

- Keyboard & Focus at App Scale
- A11y, Performance & Dynamic Content

Internationalization

TIER 1

- RTL, Locale Routing & Copy Length
- ICU, Plurals & Date/Number Formats

Design Systems (System Design)

TIER 1

- Tokens, Versioning & Governance
- Adoption, Overrides & Multi-Brand

B6.14 — HLD Case Studies & Microfrontends

24 nodes

HLD Practice Cases

TIER 1

- Practice: News Feed / Infinite Social Feed
- Practice: Google Docs Frontend
- Practice: YouTube Frontend
- Practice: Chat App (Messenger / Slack)
- Practice: Email Client (Gmail / Outlook)
- Practice: Uber Web App
- Practice: Trading Dashboard
- Practice: Pinterest Masonry
- Practice: Video Streaming (Netflix)
- Practice: E-commerce Marketplace
- Practice: Travel Booking
- Practice: Photo Sharing
- Practice: Music Streaming
- Practice: Video Conferencing
- Practice: Collaborative Spreadsheet
- Practice: Design / Drawing Tool
- Practice: Analytics Dashboard
- Practice: Kanban Board
- Practice: Live Commentary

Micro-frontends Concepts

TIER 1

- When Not to Use Microfrontends
- Integration: Build, Runtime & Routing
- Shared Dependencies & Design System

Java & Backend Engineering

Java, Spring Boot, PostgreSQL, Redis, Kafka, AWS. 255 sections · 1941 study nodes.

C1 — Core Java

16 sections

C1.1 — Syntax & Types

10 nodes

Language Syntax & Compilation	TIER 1
— Comments	
— main() & Program Entry	
— Packages	
Data Types & Variables	TIER 1
— Primitive Types	
— Reference Types	
— Type Casting	
— Local-Variable Type Inference (var)	
Scanner & User Input	TIER 2

C1.2 — Operators & Control Flow

10 nodes

Operators	TIER 1
— Arithmetic & Assignment	
— Relational & Logical	
— Bitwise Operators	
— Precedence & Associativity	
Control Flow	TIER 1
— if / else	
— switch	
— for / while / do-while	
— break & continue	

C1.3 — Arrays & Methods

9 nodes

Arrays	TIER 1
— Multidimensional Arrays	
— Arrays vs ArrayList	
Methods	TIER 1
— Parameters & Return	
— Method Overloading	
— Variable Scope	
— Recursion	
— varargs	

C1.4 — Strings & Wrappers

7 nodes

Strings	TIER 1
— String Pool	
— StringBuilder / StringBuffer	
— == vs equals() on Strings	
Wrapper Classes	TIER 1
— Autoboxing & Unboxing	

- Integer Cache

C1.5 — OOP Foundations 8 nodes

OOP	TIER 1
— Aggregation & Composition	
Encapsulation	TIER 1
Classes/Interfaces	TIER 1
— Interfaces	
— Constructors	
— this Keyword	
— Access Modifiers	

C1.6 — Inheritance & Polymorphism 9 nodes

Inheritance	TIER 1
— super Keyword	
— java.lang.Object	
Polymorphism	TIER 1
— Overloading vs Overriding	
— Dynamic vs Static Binding	
— Why static Methods Are Not Overridden	
Abstract Classes	TIER 1
— Abstract Class vs Interface	

C1.7 — Records, Enums & Nested Types 6 nodes

Records	TIER 1
Enums	TIER 1
— Enum Constructors & Fields	
Inner & Nested Classes	TIER 1
— Anonymous Classes	
— Static Nested Classes	

C1.8 — Immutability & Object Contract 5 nodes

Immutability	TIER 1
— Defensive Copies	
equals() / hashCode()	TIER 1
— hashCode()	
— equals vs Comparable	

C1.9 — Exceptions 6 nodes

Exceptions	TIER 1
— Checked vs Unchecked	
— try / catch / finally	
— try-with-resources	
— throw vs throws	
— Custom Exceptions	

C1.10 — Generics & Annotations 5 nodes

Generics	TIER 1
— Wildcards & Bounds	
— Type Erasure	
Annotations	TIER 1

C1.11 — Collections Framework

5 nodes

List / Set / Map

TIER 1

- Collection vs Collections
- Fail-Fast vs Fail-Safe
- Iterator & ListIterator
- Comparable & Comparator

C1.12 — Lists

4 nodes

ArrayList

TIER 1

- LinkedList
- Vector vs ArrayList
- List Sorting

C1.13 — Sets, Maps & Queues

9 nodes

HashMap Internals

TIER 1

- Hash Collisions & Treeify
- Resize & Transfer

HashSet

TIER 1

- TreeSet
- LinkedHashSet

TreeMap

TIER 1

PriorityQueue

TIER 1

ConcurrentHashMap

TIER 1

C1.14 — Functional Java

8 nodes

Lambdas

TIER 1

Functional Interfaces

TIER 1

- Default Methods on Interfaces

Streams

TIER 1

- Collectors
- Intermediate vs Terminal Ops

Optional

TIER 1

Method References

TIER 2

C1.15 — I/O & Serialization

7 nodes

File I/O

TIER 1

- Byte vs Character Streams
- BufferedReader / BufferedWriter
- NIO Path & Files

Serialization

TIER 1

- transient & Serializable
- serialVersionUID

C1.16 — Modern Java Language

5 nodes

Switch Expressions

TIER 1

Text Blocks

TIER 1

Sealed Classes

TIER 2

Pattern Matching

TIER 2

Module System

TIER 2

C2.1 — JDK, Runtime & JVM

4 nodes

JDK / Runtime / JVM

TIER 1

- javac, java, jar & jlink
- Modern Runtime Images vs Legacy JRE
- WORA & Platform-Specific JVM Builds

C2.2 — Bytecode & Class Files

6 nodes

Bytecode

TIER 1

- .class File Structure
- Runtime Constant Pool
- Bytecode Verification
- Inspecting Bytecode with javap
- Class Version & UnsupportedClassVersionError

C2.3 — JVM Architecture

6 nodes

JVM Architecture

TIER 1

- Class Loader Subsystem
- Runtime Data Areas
- Execution Engine

JNI & Native Libraries

TIER 2

- Native Method Stack

C2.4 — Class Loading & Initialization

8 nodes

Class Loading

TIER 1

- Loading → Linking → Initialization
- Verify, Prepare & Resolve
- Bootstrap, Platform & Application Loaders
- Parent Delegation
- <clinit>, Static Blocks & Initialization Order
- ClassNotFoundException vs NoClassDefFoundError
- Class Unloading & Loader Leaks

C2.5 — Runtime Memory Areas

7 nodes

Stack vs Heap

TIER 1

- Heap Area
- Method Area & Metaspace
- Stack Frames: Locals, Operand Stack & Frame Data
- Per-Thread PC Register
- JIT Code Cache
- Direct / Off-Heap Memory

C2.6 — Object Allocation

6 nodes

Object Allocation

TIER 1

- TLAB Fast-Path Allocation
- Object Header, Mark Word & Klass Pointer
- Compressed Oops
- Escape Analysis & Scalar Replacement
- Eden, Survivor & Promotion

C2.7 — Interpreter & JIT

7 nodes

JIT Compilation

TIER 1

- Interpreter

- Tiered Compilation: C1 & C2
- Invocation Counters & Hot Methods
- On-Stack Replacement
- Inlining & Speculative Optimization
- Deoptimization

C2.8 — Garbage Collection Foundations

10 nodes

Garbage Collection

TIER 1

- Reachability & GC Roots
- Young, Survivor & Old Generations
- Young, Major & Full GC

GC Algorithms/Concepts

TIER 1

- Mark, Sweep, Compact & Copy
- Safepoints & Stop-the-World
- Write Barriers & Remembered Sets
- Soft, Weak & Phantom References
- Finalization, Cleaner & Resource Lifecycle

C2.9 — Garbage Collectors

7 nodes

JVM Garbage Collectors

TIER 1

- Serial GC
- Parallel / Throughput GC
- G1 GC
- ZGC
- Shenandoah
- CMS (Historical / Removed)

C2.10 — Runtime Failures & Leaks

8 nodes

Common JVM Errors

TIER 1

- OutOfMemoryError: Java Heap Space
- OutOfMemoryError: Metaspace
- OutOfMemoryError: Direct Buffer Memory
- StackOverflowError

Memory Leaks

TIER 2

- Caches, Listeners & ThreadLocal Leaks
- Native Memory Leaks

C2.11 — Diagnostics & Profiling

8 nodes

JVM Profiling

TIER 2

- Java Flight Recorder & Mission Control
- async-profiler & Flame Graphs
- jcmd, jstat & jstack

Heap Dumps / Thread Dumps

TIER 2

- Thread Dumps
- Heap Dominators & Retained Size
- Unified GC Logs

C2.12 — JVM Tuning & Measurement

5 nodes

JVM Tuning Strategy

TIER 2

- -Xms, -Xmx & Heap Sizing
- Latency vs Throughput
- Container Memory Awareness
- JMH, Warm-up & Benchmark Traps

C3 — Java Concurrency

17 sections

C3.1 — Concurrency Foundations	6 nodes
Processes vs Threads	TIER 1
— Concurrency vs Parallelism	
— Platform vs Virtual Threads	
Thread Safety	TIER 1
— Shared Mutable State	
— Thread Confinement & Immutability	
C3.2 — Thread API & Lifecycle	9 nodes
Thread Lifecycle	TIER 1
— NEW, RUNNABLE, BLOCKED, WAITING & TERMINATED	
— Thread, Runnable & Callable	
— start() vs run()	
— sleep(), yield() & join()	
— Interruption & Cooperative Cancellation	
— Daemon Threads	
— UncaughtExceptionHandler	
— ThreadLocal & Cleanup	
C3.3 — Race Conditions & Safe State	6 nodes
Race Conditions	TIER 1
— Atomicity, Visibility & Ordering	
— Check-Then-Act & Read-Modify-Write	
— Critical Sections & Invariants	
— Safe Publication	
— Immutable Snapshots	
C3.4 — Java Memory Model	12 nodes
Java Memory Model	TIER 1
— Happens-Before Rules	
— Compiler & CPU Reordering	
— Sequential Consistency for Data-Race-Free Programs	
— Final-Field Semantics	
— Double-Checked Locking	
— JMM Causality & Data-Race Outcomes	
volatile	TIER 1
— Visibility & Release/Acquire Ordering	
— Why volatile Does Not Make ++ Atomic	
— Flags, Publication & Single-Writer State	
— VarHandle Memory Modes	
C3.5 — Intrinsic Monitors	7 nodes
Synchronization	TIER 1
— synchronized	
— Object Monitor & Mutual Exclusion	
— Instance vs Class Lock	
— Monitor Reentrancy	
— wait(), notify() & notifyAll()	
— Guarded Blocks & Spurious Wakeups	
C3.6 — Explicit Locks & AQS	8 nodes
Locks	TIER 1
— ReentrantLock	
— Condition	
— ReadWriteLock	
— StampedLock	
— Fair vs Non-Fair Locks	

- LockSupport park/unpark
- AbstractQueuedSynchronizer

C3.7 — Atomics & Lock-Free Programming

7 nodes

Atomic Classes

TIER 1

- Compare-And-Set (CAS)
- AtomicReference & AtomicStampedReference
- ABA Problem
- LongAdder vs AtomicLong
- Atomic Field Updaters
- Blocking, Lock-Free & Wait-Free

C3.8 — Coordination Utilities

6 nodes

Concurrency Synchronizers

TIER 1

- CountdownLatch
- CyclicBarrier
- Phaser
- Semaphore
- Exchanger

C3.9 — Concurrent Collections

8 nodes

Concurrent Collections

TIER 1

- ConcurrentHashMap
- compute, merge & putIfAbsent
- CopyOnWriteArrayList
- ConcurrentLinkedQueue / Deque
- ConcurrentSkipListMap / Set
- Weakly Consistent Iterators
- Synchronized Wrappers vs Concurrent Types

C3.10 — Producer-Consumer & Backpressure

6 nodes

Producer-Consumer

TIER 1

- BlockingQueue
- ArrayBlockingQueue vs LinkedBlockingQueue
- offer/put & poll/take
- Poison Pill & Shutdown
- Bounded Queues & Backpressure

C3.11 — Executors & Thread Pools

12 nodes

Thread Pools

TIER 1

- CPU-Bound vs I/O-Bound Sizing
- Bulkheads & Pool Isolation
- ThreadFactory, Naming & Context
- Pool Saturation Metrics
- Thread-Pool Starvation Deadlock

Executors

TIER 1

- ThreadPoolExecutor
- Work Queues: Direct, Bounded & Unbounded
- Rejection Policies & CallerRuns
- ScheduledExecutorService
- Graceful Executor Shutdown

C3.12 — Future & CompletableFuture

13 nodes

Future

TIER 1

- Callable & FutureTask
- get(), Timeout & Exception Wrapping
- Cancellation & Interruption

— ExecutorCompletionService	TIER 1
CompletableFuture	
— thenApply vs thenCompose — thenCombine, allOf & anyOf — handle, exceptionally & whenComplete — orTimeout & completeOnTimeout — Common Pool vs Explicit Executor — ThreadLocal / Trace Context Propagation — Cancellation Does Not Automatically Stop Work	
C3.13 — Fork/Join & Data Parallelism	6 nodes
ForkJoinPool	TIER 2
— Work Stealing — RecursiveTask & RecursiveAction — ForkJoin Common Pool — Parallel Streams — ManagedBlocker & Blocking Hazards	
C3.14 — Liveness & Failure Analysis	10 nodes
Deadlocks	TIER 1
— Four Coffman Conditions — Lock Ordering & Prevention — Thread Dumps & Deadlock Detection — Dining Philosophers — Timed Acquisition & Recovery	
Starvation	TIER 1
— Livelock — Priority Inversion — Fairness vs Throughput	
C3.15 — Virtual Threads	9 nodes
Virtual Threads	TIER 1
— Mounting, Unmounting & Carrier Threads — One Virtual Thread per Task — Do Not Pool — Blocking I/O with Synchronous Code — Semaphore for Resource Concurrency Limits — Why Virtual Threads Do Not Speed CPU Work — ThreadLocal Cost & Context — Pinning: JDK 21–23 vs JDK 24+ — JFR & Thread-Dump Observability	
C3.16 — Structured Concurrency & Scoped Values	11 nodes
Structured Concurrency (Preview)	TIER 2
— StructuredTaskScope — Parent–Child Task Trees — Failure, Cancellation & Join Policies — Joiners & Fan-Out/Fan-In — JDK 26 Preview Status & API Evolution — Structured Thread Dumps	
Scoped Values	TIER 2
— ScopedValue vs ThreadLocal — Inheritance into Structured Subtasks — Immutable Request Context	
C3.17 — Concurrency Testing	6 nodes
Testing Concurrent Code	TIER 2
— Deterministic Tests with Latches & Barriers	

- Why Tests Must Not Coordinate with sleep()
- JCTest & JMM Litmus Tests
- Lincheck for Concurrent Data Structures
- Stress Loops, Timeouts & Failure Capture

C4 — Networking & Web

19 sections

C4.1 — Network Foundations & Models

10 nodes

Computer Networking Fundamentals

TIER 1

- Why Networks Exist
- LAN, WAN, PAN & Internet
- Client–Server vs Peer-to-Peer
- Bandwidth, Throughput, Latency & Jitter

TCP/IP Concepts

TIER 1

- OSI Seven-Layer Model
- TCP/IP Model
- OSI vs TCP/IP
- Encapsulation: Data, Segment, Packet & Frame

C4.2 — Devices, Media & Wireless

11 nodes

Network Devices

TIER 2

- Network Interface Card & MAC Address
- Hub vs Switch vs Router
- Modem, Gateway & Wireless Access Point
- Physical Layer & Signaling
- Copper, Fiber & Radio Media
- Simplex, Half-Duplex & Full-Duplex

Wireless Networks

TIER 3

- Wi-Fi & IEEE 802.11
- Bluetooth & Short-Range Networking
- Channels, Interference & Roaming

C4.3 — Ethernet & Data Link Layer

8 nodes

Ethernet & Data Link Layer

TIER 2

- Ethernet Frame & MTU
- MAC Addressing & Broadcast Domains
- Layer-2 Switching & MAC Tables
- VLANs & 802.1Q Tags
- Framing
- CRC & Error Detection
- Link-Layer Flow Control & ARQ

C4.4 — IP Addressing & Subnetting

9 nodes

Internet Protocol & Addressing

TIER 1

- IPv4 Address Structure
- IPv4 Datagram Header
- CIDR & Subnet Masks
- Network, Broadcast & Host Ranges
- VLSM & Supernetting
- Private, Public, Loopback & Link-Local IPs
- IPv4 vs IPv6
- IPv6 Addressing & Neighbor Discovery

C4.5 — Routing, NAT & Network Control

9 nodes

Routing & Network Boundaries

TIER 1

- Default Gateway & Routing Table

- Static vs Dynamic Routing
- Distance-Vector vs Link-State Routing
- OSPF & BGP Awareness
- NAT, PAT & Port Exhaustion
- ARP vs IPv6 NDP
- DHCP Lease Lifecycle
- ICMP, Errors & Reachability

C4.6 — Transport Layer & Sockets

8 nodes

Transport Layer

TIER 1

- Ports, Sockets & the Five-Tuple
- TCP vs UDP
- UDP Datagrams, Loss & Ordering
- bind, listen, accept & connect
- Byte Streams vs Message Boundaries
- SCTP Awareness
- QUIC Transport Overview

C4.7 — TCP Reliability & Performance

10 nodes

Transmission Control Protocol

TIER 1

- TCP Three-Way Handshake
- FIN, RST & Connection Termination
- Sequence Numbers, ACKs & Retransmission
- Receive Window & Flow Control
- Slow Start & Congestion Control
- TIME_WAIT & Ephemeral-Port Exhaustion
- SYN Queue, Accept Queue & Backlog
- Nagle, Delayed ACK & TCP_NODELAY
- MTU, MSS & Path MTU Discovery

C4.8 — DNS & Service Discovery

10 nodes

Domain Name System

TIER 1

- Root, TLD & Authoritative Servers
- Recursive vs Iterative Resolution
- A, AAAA, CNAME, MX, TXT, NS & SRV
- Caching, TTL & Negative Answers
- UDP, TCP, DoT, DoH & DoQ
- DNSSEC & Cache-Poisoning Defense
- SVCB & HTTPS Records
- Service Discovery & Split-Horizon DNS
- DNS Failure, Stale Cache & Failover

C4.9 — TLS & HTTPS

10 nodes

Transport Layer Security

TIER 1

- HTTPS
- TLS 1.3 Handshake
- Certificates, PKI & Chain Validation
- SNI & ALPN
- Forward Secrecy & AEAD
- Session Resumption & Replay-Safe 0-RTT
- Mutual TLS
- TLS Termination & Re-Encryption
- Certificate Rotation, Revocation & OCSP Stapling

C4.10 — HTTP Semantics

10 nodes

HTTP

TIER 1

- Request & Response Message Structure
- HTTP Methods

- Safe, Idempotent & Cacheable Methods
- HTTP Headers
- HTTP Status Codes
- URI, URL, Authority & Origin
- Media Types, MIME & Content-Type
- Content Negotiation
- Retries, Idempotency Keys & Duplicate Effects

C4.11 — HTTP Connections, Caching & Delivery

11 nodes

Connection Reuse & Keep-Alive

TIER 1

- Connection Pooling
- HTTP/1.1 Pipelining & Head-of-Line Blocking
- Content-Length, Chunked Transfer & Trailers
- Streaming & Range Requests

HTTP Caching

TIER 1

- Cache-Control Directives
- ETag, Last-Modified & Conditional Requests
- Public, Private, no-cache & no-store
- Vary & Cache-Key Correctness
- gzip, Brotli & Compression Trade-Offs

C4.12 — HTTP/2, HTTP/3 & QUIC

11 nodes

HTTP/2

TIER 2

- Binary Frames & Streams
- Multiplexing & TCP Head-of-Line Blocking
- HPACK Header Compression
- Stream & Connection Flow Control
- Priorities & Why Server Push Declined

HTTP/3

TIER 2

- QUIC Streams & Loss Isolation
- QPACK Header Compression
- Connection IDs & Migration
- Alt-Svc, HTTPS Records, ALPN & Fallback

C4.13 — Cookies, Sessions & Browser Boundaries

12 nodes

Cookies

TIER 1

- Domain, Path, Expires & Max-Age
- Secure, HttpOnly & SameSite
- Cookie Scope, __Host- & __Secure- Prefixes

Sessions

TIER 1

- Session IDs & Server-Side Stores
- Sticky vs Shared Sessions
- Session Rotation & Fixation Defense

CORS

TIER 1

- Same-Origin Policy
- Simple Requests & Preflight
- Credentials, Origins & Cache Variation

C4.14 — Proxies, Load Balancers & CDN

15 nodes

Reverse Proxies

TIER 1

- Forward vs Reverse Proxy
- Forwarded & X-Forwarded-* Headers
- Buffering, Timeouts & Request Limits
- API Gateway
- Gateway Authentication, Rate Limits & Routing

Load Balancers

TIER 1

- Layer 4 vs Layer 7 Load Balancing
- Round Robin, Least Connections & Hashing

— Health Checks, Readiness & Connection Draining — Session Affinity & Long-Lived Connections	TIER 1
Content Delivery Networks	
— Edge Caching & Origin Shield — Invalidation, Purge & Versioned Assets — Anycast & Geo-Routing	
C4.15 — API & Realtime Protocols	18 nodes
REST	TIER 1
— Resources, Representations & Uniform Interface — Pagination, Errors & Versioning — JSON — RPC vs REST — OpenAPI & Swagger Contracts — Problem Details Error Responses — GraphQL over HTTP	
gRPC	TIER 2
— Protocol Buffers & Schema Evolution — Unary, Client, Server & Bidirectional Streaming — Deadlines, Cancellation & Status Codes	
Realtime Web Protocols	TIER 1
— WebSocket Handshake, Frames & Heartbeats — Server-Sent Events — Long Polling — Webhooks, Signatures & Retries — WebTransport	
C4.16 — Network Security Fundamentals	9 nodes
Network Security Fundamentals	TIER 1
— Authentication vs Authorization — Encryption in Transit — Firewalls & Stateful Filtering — Segmentation, DMZ & Zero Trust — VPN, Tunnels & IPsec — IDS & IPS — DDoS, SYN Floods & Rate Limiting — MAC Filtering Limitations	
C4.17 — Cloud & Container Networking	8 nodes
Cloud Networking	TIER 2
— VPCs & Public/Private Subnets — Internet Gateway vs NAT Gateway — Security Groups vs Network ACLs — Cloud DNS & Service Discovery — Container Ports, Bridges & Overlay Networks — Kubernetes Pod, Service & Ingress Networking — Service Mesh & Sidecar Proxies	
C4.18 — Performance & Troubleshooting	15 nodes
Network Troubleshooting	TIER 1
— What Happens When You Enter a URL? — ping, ICMP & traceroute — dig, nslookup & DNS Diagnosis — curl & openssl s_client — ss, netstat & lsof — tcpdump & Wireshark — Layered Connectivity Diagnosis	
Network Performance	TIER 1

- RTT & End-to-End Latency Budgets
- Packet Loss, Retransmission & Tail Latency
- Timeouts, Retries, Backoff & Jitter
- Bandwidth–Delay Product
- QoS, Token Bucket & Leaky Bucket
- DNS, Connect, TLS, TTFB & Download Timing

C4.19 — Java Networking APIs

9 nodes

Java Networking APIs

TIER 1

- URI, URL & InetAddress
- Java 11+ HttpClient
- Connect, Request & Read Timeouts
- sendAsync & CompletableFuture
- Connection Reuse, HTTP/2 & Pooling
- RestClient, WebClient & HTTP Interfaces
- ServerSocket, Channels & Selectors
- Netty Event Loop & Channels

C5 — Spring Core

14 sections

C5.1 — IoC & the Container

5 nodes

IoC

TIER 1

- Hollywood Principle
- IoC vs Dependency Injection
- IoC vs Service Locator
- XML, Annotations & Java Config Metadata

C5.2 — ApplicationContext

6 nodes

ApplicationContext

TIER 1

- BeanFactory vs ApplicationContext
- refresh(), start & close
- WebApplicationContext
- getBean() as a Service Locator
- ApplicationContextAware

C5.3 — Dependency Injection

8 nodes

Dependency Injection

TIER 1

- Constructor Injection
- Setter Injection
- Field Injection
- @Autowired, @Qualifier & @Primary
- @Inject vs @Resource
- ObjectProvider & Delayed Lookup
- Optional Dependencies

C5.4 — Beans, Names & Scopes

8 nodes

Beans

TIER 1

- @Component, @Service, @Repository, @Controller
- Bean Names & Aliases
- Singleton Scope & Thread Safety
- Prototype Scope
- Request, Session & Application Scopes
- FactoryBean

- @Lazy Initialization

C5.5 — Bean Lifecycle

8 nodes

Bean Lifecycle

TIER 1

- Instantiation & Population
- @PostConstruct & @PreDestroy
- InitializingBean & DisposableBean
- init-method & destroy-method
- BeanPostProcessor in the Lifecycle
- Prototype Destroy Ownership
- SmartLifecycle & Phased Start

C5.6 — Component Scanning

5 nodes

Component Scanning

TIER 1

- Base Packages & Default-Package Trap
- includeFilters & excludeFilters
- Stereotype Semantics
- spring.components Index

C5.7 — Java Configuration

6 nodes

Configuration

TIER 1

- @Bean Factory Methods
- Full @Configuration vs Lite @Bean
- @Import & @ImportResource
- @Conditional Bean Registration
- XML Configuration (Legacy)

C5.8 — Profiles & Environment

5 nodes

Profiles

TIER 1

- Active & Default Profiles
- Environment Abstraction
- @Value & @PropertySource
- SpEL in Injection Points

C5.9 — Wiring Conflicts & Cycles

6 nodes

Wiring Conflicts & Circular Dependencies

TIER 1

- NoUniqueBeanDefinitionException
- Circular Dependencies
- allow-circular-references (Boot 2.6+)
- @Lookup Method Injection
- @DependsOn

C5.10 — Events, Resources & i18n

5 nodes

Events, Resources & MessageSource

TIER 1

- ApplicationEvent & ApplicationPublisher
- @EventListener & Transactional Events
- Resource & ResourceLoader
- MessageSource & i18n

C5.11 — Aspect-Oriented Programming

7 nodes

AOP

TIER 2

- Join Points & Pointcuts
- Before, After, AfterReturning, AfterThrowing
- @Around & ProceedingJoinPoint
- Aspect @Order
- Spring AOP vs AspectJ Weaving

- @Transactional as AOP

C5.12 — Spring Proxies

7 nodes

Spring Proxies

TIER 2

- JDK Dynamic Proxy vs CGLIB
- Self-Invocation Bypass
- AopContext.exposeProxy
- Scoped Proxies
- proxy-target-class
- final Classes, Methods & Visibility

C5.13 — Container Internals

5 nodes

BeanFactory Internals

TIER 2

- BeanDefinition
- BeanFactoryPostProcessor
- InstantiationAwareBeanPostProcessor
- ConfigurationClassPostProcessor

C5.14 — Testing Spring Core

5 nodes

Testing Spring Core

TIER 1

- Plain Unit Tests via Constructor DI
- @ContextConfiguration
- @DirtiesContext
- @MockitoBean vs @MockBean

C6 — Spring Boot

20 sections

C6.1 — Boot Foundations & Project Structure

9 nodes

Spring Boot Foundations & Project Structure

TIER 1

- Spring Framework vs Spring Boot
- Spring Initializr
- Package by Layer vs Package by Feature
- Maven & Gradle Boot Plugins
- Starter Dependencies
- BOM & Dependency Management
- Executable & Layered JARs
- DevTools & Restart Classloader

C6.2 — SpringApplication & Startup

8 nodes

SpringApplication & Startup Lifecycle

TIER 1

- @SpringBootApplication Composition
- SpringApplication.run()
- Servlet, Reactive & Non-Web Application Types
- Startup Events & Listeners
- CommandLineRunner & ApplicationRunner
- Application Exit Codes
- FailureAnalyzers & Startup Diagnostics

C6.3 — Auto-configuration & Starters

10 nodes

Auto-configuration

TIER 1

- @EnableAutoConfiguration
- AutoConfiguration.imports
- @ConditionalOn* Annotations
- Back-off & User Overrides
- Auto-configuration Ordering

- Condition Evaluation Report
- Excluding Auto-configuration
- Custom Auto-configuration & Starters
- Boot 4 Modular Auto-configuration

C6.4 — Externalized Configuration

12 nodes

Externalized Configuration & Properties

TIER 1

- Property Source Precedence
- Config Data API & Imports
- application.properties vs YAML
- Environment, CLI & SPRING_APPLICATION_JSON
- Profiles, Groups & Activation
- @ConfigurationProperties
- @Value vs @ConfigurationProperties
- Relaxed Binding
- Duration, DataSize & Custom Conversion
- Configuration Validation
- configtree: & Mounted Secrets

C6.5 — Embedded Web Runtime

8 nodes

Embedded Web Server & Runtime

TIER 1

- Embedded Tomcat
- Jetty & Undertow Alternatives
- Server Port, Context Path & Compression
- HTTP/2, TLS & SSL Bundles
- Servlet Thread-per-Request Model
- Boot Virtual Thread Integration
- Executable JAR vs Traditional WAR

C6.6 — Spring MVC Request Lifecycle

10 nodes

Request/Response Lifecycle

TIER 1

- Servlet API & Container Bridge
- DispatcherServlet Front Controller
- HandlerMapping & HandlerAdapter
- HandlerMethodArgumentResolver
- Return Value Handlers
- HttpMessageConverters
- Content Negotiation
- MVC Async: Callable, DeferredResult & Streaming
- Spring MVC vs WebFlux

C6.7 — REST Controllers & HTTP Semantics

9 nodes

REST Controllers

TIER 1

- @Controller vs @RestController
- @RequestMapping & HTTP Method Mappings
- Path, Query & Header Binding
- @RequestBody, @ResponseStatus & ResponseEntity
- HTTP Status & Method Semantics
- MVC CORS Configuration
- Multipart Uploads & Downloads
- OpenAPI with springdoc

C6.8 — DTOs, JSON & API Boundaries

9 nodes

DTOs & API Boundaries

TIER 1

- Request vs Response DTOs
- Java Records as DTOs
- Why APIs Must Not Expose Entities
- Jackson Serialization & Deserialization

- Naming, Inclusion & Unknown Fields
- ObjectMapper & JsonMapper Customization
- Jackson 2 to Jackson 3 Migration
- Manual Mapping vs MapStruct

C6.9 — Validation & Data Binding

9 nodes

Validation & Data Binding

TIER 1

- Jakarta Bean Validation
- @Valid vs @Validated
- Built-in Constraint Annotations
- Custom Constraints
- Validation Groups
- Method Validation
- BindingResult & Field Errors
- Converter, Formatter & DataBinder

C6.10 — Exception Handling & Error Contracts

8 nodes

Global Exception Handling

TIER 1

- @ExceptionHandler
- @ControllerAdvice & @RestControllerAdvice
- HandlerExceptionResolver Chain
- ProblemDetail & RFC 9457
- Stable Validation Error Responses
- Correlation IDs in Error Responses
- Preventing Error Information Leaks

C6.11 — Filters, Interceptors & Web Hooks

8 nodes

Servlet Filters & MVC Interceptors

TIER 1

- HandlerInterceptor
- Filter vs Interceptor vs AOP
- OncePerRequestFilter
- FilterRegistrationBean & Ordering
- Request/Response Wrappers & Body Caching
- WebMvcConfigurer
- Controller Advice Binding Hooks

C6.12 — API Versioning & Evolution

7 nodes

API Versioning & Evolution

TIER 2

- URI, Header & Media-Type Versioning
- Spring Framework 7 API Versioning
- Breaking vs Additive Changes
- Deprecation, Sunset & Migration Windows
- Versioned Controllers & DTOs
- Compatibility & Contract Tests

C6.13 — Pagination, Sorting & Collection APIs

8 nodes

Pagination, Sorting & Collection APIs

TIER 1

- Pageable, Page & Sort
- Page vs Slice
- Offset Pagination
- Keyset/Cursor Pagination
- Bounds, Maximum Page Size & Stable Ordering
- Metadata, Links & HATEOAS
- Filtering & Search Parameters

C6.14 — HTTP Clients & Service Integration

9 nodes

HTTP Clients & Service Integration

TIER 1

- RestClient
- WebClient
- HTTP Service Clients & @HttpExchange
- RestTemplate (Legacy)
- Base URL, Headers & Message Conversion
- Connect, Read & Request Timeouts
- Status Handling & Error Mapping
- Client Observations & Context Propagation

C6.15 — Task Execution & Scheduling

9 nodes

Task Execution & Scheduling

TIER 2

- @EnableAsync & @Async
- @Async Proxy & Self-Invocation
- @EnableScheduling & @Scheduled
- Cron, fixedDelay & fixedRate
- TaskExecutor Auto-configuration
- TaskScheduler Auto-configuration
- TaskDecorator & Context Propagation
- Duplicate Scheduling in a Cluster

C6.16 — Logging & Diagnostics

8 nodes

Logging & Diagnostics

TIER 1

- SLF4J, Logback & LoggingSystem
- Levels, Groups & Package Configuration
- Console Patterns, Files & Rotation
- Built-in Structured Logging
- MDC & Correlation IDs
- Runtime Log-Level Changes
- --debug & Startup Reports

C6.17 — Actuator & Application Availability

9 nodes

Actuator & Application Availability

TIER 1

- Actuator Endpoint Model
- Exposure, Access & Management Port
- HealthContributor & HealthIndicator
- Liveness, Readiness & Availability States
- Health Groups
- Micrometer Metrics Bridge
- Securing Sensitive Endpoints
- Custom Actuator Endpoints

C6.18 — Spring Boot Testing

11 nodes

Spring Boot Testing

TIER 1

- @SpringBootTest
- @WebMvcTest & MockMvc
- @WebFluxTest & WebTestClient
- @JsonTest & REST Client Tests
- Test Slices & Restricted Contexts
- @MockitoBean & @MockitoSpyBean
- Context Caching & @DirtiesContext
- Testcontainers & @ServiceConnection
- @DynamicPropertySource
- RANDOM_PORT & Real HTTP Tests

C6.19 — AOT, Native Images & Runtime Efficiency

8 nodes

AOT Processing & Native Images

TIER 2

- Spring AOT Processing
- GraalVM Native Images

- RuntimeHints & Reflection Metadata
- Generated Initializers & Proxies
- Native Build & Test Lifecycle
- Startup/Memory vs Build/Compatibility Trade-offs
- CRaC Checkpoint/Restore

C6.20 — Production Lifecycle & Packaging

8 nodes

Production Lifecycle & Packaging

TIER 1

- Graceful Shutdown & Timeout
- spring.main.keep-alive & Daemon Threads
- OCI Images with Cloud Native Buildpacks
- Development-time Docker Compose Support
- Layered Container Images
- Remote Admin & Unsafe Endpoint Traps
- Production Configuration Checklist

C7 — SQL & PostgreSQL

20 sections

C7.1 — Relational & PostgreSQL Foundations

12 nodes

Relational Databases & PostgreSQL

TIER 1

- What SQL Is
- RDBMS vs Files & Spreadsheets
- Relations, Tuples, Attributes & Domains
- DDL, DML, DQL, DCL & TCL
- Standard SQL vs PostgreSQL Dialect
- PostgreSQL Client/Server Architecture
- Installation, Cluster & Database Initialization
- Getting Started & Demo Database
- psql Essentials
- pgAdmin 4 & Database GUI Tools
- Database, Schema & search_path

C7.2 — SQL Syntax, Types & Expressions

16 nodes

SQL Syntax, Data Types & Expressions

TIER 1

- Identifiers, Keywords & Quoting
- Statements, Semicolons & Comments
- Numeric, Boolean & Character Types
- DATE, TIME, TIMESTAMP & INTERVAL
- timestamp vs timestamptz
- NULL & Three-Valued Logic
- SQL Operators
- Arithmetic & Comparison Operators
- AND, OR & NOT
- AND
- OR
- NOT
- String Concatenation
- CAST, :: & Type Conversion
- Operator Precedence

C7.3 — Databases, Schemas & Table DDL

17 nodes

Database, Schema & Table DDL

TIER 1

- CREATE & DROP DATABASE
- CREATE DATABASE
- DROP DATABASE
- CREATE & DROP SCHEMA
- CREATE TABLE
- ALTER TABLE

- ADD COLUMN
- ALTER COLUMN
- DROP COLUMN
- DROP vs TRUNCATE vs DELETE
- CREATE INDEX Syntax
- Renaming Tables & Columns
- Temporary & Unlogged Tables
- CREATE TABLE AS & SELECT INTO
- COPY for Bulk Import/Export
- Transactional DDL in PostgreSQL

C7.4 — Keys, Constraints & Generated Values

16 nodes

Keys, Constraints & Generated Values

TIER 1

- PRIMARY KEY
- FOREIGN KEY & Referential Actions
- UNIQUE & NULL Semantics
- NOT NULL
- CHECK Constraints
- DEFAULT Values
- Composite & Candidate Keys
- Named Table & Column Constraints
- DROP CONSTRAINT
- IDENTITY Columns
- AUTO INCREMENT vs IDENTITY
- Sequences & nextval()
- serial / bigserial (Legacy Convenience)
- Deferrable Constraints
- Exclusion Constraints

C7.5 — CRUD & Data Modification

15 nodes

SELECT, INSERT, UPDATE & DELETE

TIER 1

- SELECT & Projection
- SELECT * vs Explicit Columns
- Fetching Rows
- DISTINCT & DISTINCT ON
- INSERT & Multi-row VALUES
- NULL Values in INSERT/UPDATE
- INSERT INTO ... SELECT
- UPDATE & UPDATE ... FROM
- DELETE & DELETE ... USING
- RETURNING
- INSERT ... ON CONFLICT (UPSERT)
- MERGE
- Data-Modifying CTEs
- Safe UPDATE/DELETE Practices

C7.6 — Filtering, Ordering & Conditional Logic

17 nodes

Filtering, Ordering & Conditional Logic

TIER 1

- WHERE
- LIKE, ILIKE & Wildcards
- SQL Wildcards
- IN, NOT IN & NULL Trap
- BETWEEN & Range Boundaries
- IS NULL, IS DISTINCT FROM
- EXISTS & NOT EXISTS
- ANY, ALL & Array Comparisons
- ANY / SOME
- ALL
- CASE Expressions
- COALESCE & NULLIF
- ORDER BY & NULLS FIRST/LAST

- SELECT TOP vs LIMIT & FETCH
- LIMIT, OFFSET & FETCH
- Column & Table Aliases

C7.7 — Functions, Aggregation & Grouping

17 nodes

- Functions, Aggregation & Grouping
- COUNT, SUM & AVG
 - COUNT()
 - SUM()
 - AVG()
 - MIN & MAX
 - MIN()
 - MAX()
 - GROUP BY
 - HAVING vs WHERE
 - Aggregate NULL Semantics
 - FILTER on Aggregates
 - String Functions
 - Numeric Functions
 - Date/Time Functions & date_trunc
 - Regular Expressions
 - GROUPING SETS, ROLLUP & CUBE

TIER 1

C7.8 — Joins & Set Operations

15 nodes

- Joins & Set Operations
- INNER JOIN
 - LEFT OUTER JOIN
 - RIGHT & FULL OUTER JOIN
 - RIGHT JOIN
 - FULL JOIN
 - CROSS JOIN
 - Self Join
 - ON vs WHERE in Outer Joins
 - Join Cardinality & Duplicate Rows
 - LATERAL Joins
 - UNION vs UNION ALL
 - UNION ALL
 - INTERSECT & EXCEPT
 - Set Operation Type Compatibility

TIER 1

C7.9 — Subqueries & CTEs

11 nodes

- Subqueries
- Scalar Subqueries
 - Multi-row Subqueries
 - Correlated Subqueries
 - EXISTS as Semi/Anti Join
 - Subquery vs Join
- Common Table Expressions
- WITH Queries
 - WITH RECURSIVE
 - Recursive CTE Cycle Handling
 - CTE Inlining & MATERIALIZED

TIER 1

TIER 1

C7.10 — Window Functions & Advanced Querying

11 nodes

- Window Functions
- OVER, PARTITION BY & ORDER BY
 - ROW_NUMBER, RANK & DENSE_RANK
 - LAG & LEAD
 - FIRST_VALUE, LAST_VALUE & NTH_VALUE

TIER 1

- Running & Moving Aggregates
- ROWS, RANGE & GROUPS Frames
- Named Windows
- Top-N per Group
- Gaps & Islands
- Pivoting with FILTER/crosstab

C7.11 — Data Modeling & Normalization

15 nodes

Relational Schema Design

TIER 1

- Entities, Relationships & ER Diagrams
- Cardinality & Optionality
- One-to-One, One-to-Many & Many-to-Many
- Keys in the Relational Model
- Superkeys & Candidate Keys
- Natural vs Surrogate Keys
- Functional Dependencies

Normalization

TIER 1

- First Normal Form
- Second Normal Form
- Third Normal Form
- Boyce–Codd Normal Form
- Insert, Update & Delete Anomalies
- Denormalization & Trade-offs

C7.12 — Transactions, ACID & Isolation

16 nodes

Transactions

TIER 1

- BEGIN, COMMIT & ROLLBACK
- Autocommit
- SAVEPOINT & Partial Rollback
- Transaction Boundary Design

ACID

TIER 1

- Atomicity
- Consistency
- Isolation
- Durability & WAL

PostgreSQL Isolation Levels

TIER 1

- Read Committed
- Repeatable Read / Snapshot Isolation
- Serializable Snapshot Isolation
- Dirty, Nonrepeatable, Phantom & Serialization Anomalies
- Retrying Serialization Failures

C7.13 — MVCC, Locks & Deadlocks

14 nodes

MVCC

TIER 1

- Tuple Versions: xmin, xmax & Snapshots
- Readers, Writers & Visibility
- Dead Tuples & Table Bloat
- Long Transactions & Snapshot Retention

Locks

TIER 1

- Table Lock Modes
- Row Locks & SELECT FOR UPDATE
- NOWAIT & SKIP LOCKED
- Advisory Locks
- pg_locks & Blocking Queries

Deadlocks

TIER 1

- Deadlock Detection & Victim Abort

- Consistent Lock Ordering

C7.14 — Indexes & Access Methods

15 nodes

PostgreSQL Indexes

TIER 1

- Selectivity & Cardinality
- Unique Indexes
- Covering Indexes & INCLUDE
- Partial Indexes
- Expression Indexes
- GIN Indexes
- GiST & SP-GiST Indexes
- BRIN Indexes
- Hash Indexes
- Index Write, Storage & Vacuum Cost
- CREATE INDEX CONCURRENTLY

B-Tree / B+ Tree Concepts

TIER 1

Composite Indexes

TIER 1

- Column Order & Leftmost Prefix

C7.15 — Query Planning & Performance

14 nodes

Query Plans

TIER 1

- Sequential, Index & Bitmap Scans
- Nested Loop, Hash & Merge Joins
- Sort, Aggregate & Materialize Nodes
- Planner Costs, Rows & Width
- Statistics & ANALYZE
- Extended Statistics

EXPLAIN & EXPLAIN ANALYZE

TIER 1

- BUFFERS, WAL, SETTINGS & Timing

Query Optimization

TIER 1

- Sargability
- Query Rewriting & Predicate Placement
- work_mem, Sort/Hash Spill
- pg_stat_statements

C7.16 — PostgreSQL Types & Document Features

12 nodes

PostgreSQL Advanced Types

TIER 1

- UUID
- ENUM & DOMAIN
- Arrays
- Range & Multirange Types
- Network Address Types
- Geometric Types
- JSON vs JSONB
- JSONB Operators & SQL/JSON
- JSONB Indexing with GIN
- Full-Text Search: tsvector & tsquery
- Generated Columns

C7.17 — Views, Functions, Procedures & Triggers

12 nodes

Views & Database Programmability

TIER 2

- Views
- Updatable Views & CHECK OPTION
- Materialized Views & REFRESH
- SQL Functions
- PL/pgSQL
- Functions vs Procedures
- Stored Procedures & CALL

- Routine Parameters IN/OUT
- Row & Statement Triggers
- BEFORE, AFTER & INSTEAD OF
- Trigger Trade-offs & Hidden Side Effects

C7.18 — Connectivity, Security & Data Safety

18 nodes

Connections & Pooling

TIER 1

- PostgreSQL Wire Protocol Overview
- max_connections & Backend Cost
- HikariCP vs PgBouncer
- Session vs Transaction Pooling
- SQL Injection Defense
- SQL Parameters
- Prepared Statements & Parameters
- Data Encryption
- pgcrypto
- PostgreSQL Hosting & Managed Services
- Roles, GRANT, REVOKE & Least Privilege
- Row-Level Security
- TLS Connections & Certificate Verification
- Logical vs Physical Backup & Restore
- pg_dump, pg_restore & pg_basebackup
- WAL Archiving & Point-in-Time Recovery
- Versioned Schema Migrations

C7.19 — Partitioning & Large Tables

9 nodes

Table Partitioning

TIER 2

- Declarative Partitioning
- Range, List & Hash Partitioning
- Choosing a Partition Key
- Partition Pruning
- Tuple Routing & Default Partitions
- Partitioned Indexes & Constraints
- Attach, Detach & Retention
- Partitioning Trade-offs

C7.20 — Replication, Vacuum & Operations

13 nodes

PostgreSQL Replication

TIER 2

- Physical Streaming Replication
- Write-Ahead Log (WAL)
- Synchronous vs Asynchronous Replication
- Replication Slots & WAL Retention
- Logical Replication & Publications
- Read Replicas & Replication Lag
- VACUUM & Dead Tuple Reclamation
- Autovacuum Tuning
- Transaction ID Freeze & Wraparound
- REINDEX, CLUSTER & pg_repack
- System Catalogs & Statistics Views
- Monitoring, Capacity & Slow-Query Workflow

C8 — JPA / Hibernate

16 sections

C8.1 — ORM Foundations & Technology Boundaries

8 nodes

JPA, Hibernate & Spring Data JPA

TIER 1

- JPA as a Specification
- Hibernate as JPA Provider
- Spring Data JPA Module

- ORM vs JDBC / jOOQ / MyBatis
- EntityManager vs Hibernate Session
- JPQL vs HQL
- javax.persistence vs jakarta.persistence

C8.2 — Entities, Identity & Mapping Basics

8 nodes

Entities

TIER 1

- Entity Requirements & No-arg Constructor
- @Table, @Column & Naming Strategies
- Field Access vs Property Access
- Constructors, Setters & Domain Methods
- equals/hashCode for Entities
- toString(), Lombok & Lazy Loading Traps
- Mutable Entities vs Immutable Domain Models

C8.3 — Identifiers, Generated Values & Value Types

9 nodes

Identifiers & Generated Values

TIER 1

- @Id & @GeneratedValue
- IDENTITY vs SEQUENCE Strategies
- Sequence allocationSize & Insert Throughput
- UUID Identifiers
- Composite IDs with @EmbeddedId / @IdClass
- @Embeddable Value Objects
- Enum Mapping: ORDINAL vs STRING
- AttributeConverter

C8.4 — Relationships, Ownership & Cascades

9 nodes

Relationships

TIER 1

- @ManyToOne
- @OneToMany
- @OneToOne
- @ManyToMany and Join Tables
- Owning Side, mappedBy & Foreign Keys
- Cascade Types
- orphanRemoval
- Bidirectional Association Synchronization

C8.5 — Persistence Context, Lifecycle & Flush

8 nodes

Persistence Context

TIER 1

- First-level Cache / Identity Map
- Transient, Managed, Detached & Removed
- persist(), merge(), remove()
- Flush Modes and Query-time Flush
- clear(), detach() & refresh()
- Open Session in View

Entity Lifecycle

TIER 1

C8.6 — Dirty Checking, Write-behind & SQL Generation

7 nodes

Dirty Checking

TIER 1

- Loaded State Snapshots
- Transactional Write-behind
- save() vs Managed Entity Changes
- @DynamicUpdate Trade-offs
- Bulk Updates Bypass Persistence Context

C8.7 — Fetching, Lazy Loading & N+1 Control

10 nodes

Lazy/Eager Loading

TIER 1

- Eager Loading
- Hibernate Proxies and Persistent Collections
- LazyInitializationException

N+1 Problem

TIER 1

Fetch Joins

TIER 1

- @EntityGraph
- Batch Fetching and default_batch_fetch_size
- SUBSELECT Fetching
- MultipleBagFetchException

C8.8 — Repositories, Queries & Projections

9 nodes

Repositories

TIER 1

- CrudRepository vs JpaRepository
- Derived Query Methods
- @Query with JPQL and Native SQL
- Named and Positional Query Parameters
- JpaRepositoryExecutor / Criteria API
- Query by Example
- Interface, DTO and Record Projections
- Page, Slice, Pageable and Sort

C8.9 — Transactions, Locking & Concurrency

9 nodes

JPA Transactions

TIER 1

- @Transactional Boundaries
- Propagation and Isolation with JPA
- readOnly Transactions
- Rollback Rules and Checked Exceptions

Optimistic/Pessimistic Locking

TIER 1

- @Version Column
- OptimisticLockException and Retry Design
- Pessimistic Locking

C8.10 — Schema, Validation & Migrations

8 nodes

Schema Management with JPA and Migrations

TIER 1

- hibernate.hbm2ddl.auto / ddl-auto
- Flyway / Liquibase with JPA
- @Column nullable, unique and length
- Bean Validation on Entities
- Unique Constraints and Index Metadata
- CreatedDate, LastModifiedDate and Auditing
- Soft Delete Patterns

C8.11 — Caching, Performance & Observability

8 nodes

Hibernate Performance and Observability

TIER 1

- Second-level Cache
- Query Cache
- Hibernate Statistics
- SQL Logging, p6spy and datasource-proxy
- Fetch Size and Streaming Results
- Batch Inserts and Updates

- Read-only Queries and Hints

C8.12 — Inheritance, Polymorphism & Advanced Mapping

8 nodes

Advanced Hibernate Mapping

TIER 1

- Inheritance Mapping Strategies
- @MappedSuperclass
- SINGLE_TABLE Strategy
- JOINED Strategy
- TABLE_PER_CLASS Strategy
- @SecondaryTable
- @Filter, @Where and Tenant Filters

C8.13 — Spring Data JPA Internals & Extension Points

7 nodes

Spring Data JPA Internals

TIER 1

- Repository Proxies
- Query Lookup Strategy
- Custom Repository Implementations
- Domain Events from Aggregates
- Repository Transaction Defaults
- Entity Callbacks and Listeners

C8.14 — Testing Persistence

7 nodes

Testing JPA and Hibernate

TIER 1

- @DataJpaTest
- TestEntityManager
- PostgreSQL Testcontainers
- H2 Dialect Traps
- Repository Query Tests
- N+1 Regression Tests

C8.15 — API Boundaries, DTOs & Production Patterns

7 nodes

Avoiding ORM Abuse

TIER 1

- DTO Boundaries and Entity Exposure
- Aggregate Boundaries
- Transaction Script vs Rich Domain
- Jackson Serialization of Entities
- OSIV Production Trade-off
- When to Use Native SQL or JDBC

C8.16 — Interview Scenarios & Troubleshooting

7 nodes

JPA/Hibernate Troubleshooting

TIER 1

- Debugging Generated SQL
- detached entity passed to persist
- TransientObjectException
- StaleObjectStateException
- Connection Leak Symptoms
- Production Query Spike Investigation

C9 — Security

15 sections

C9.1 — Authentication vs Authorization

6 nodes

Authentication vs Authorization

TIER 1

- Identification vs Authentication
- Authorization After Authentication
- Fail-Closed Defaults

- Session vs Token Authentication Models
- MFA & Step-up Authentication

C9.2 — Sessions, Cookies & Browser Storage

6 nodes

Sessions (Security)

TIER 1

- Session Fixation
- HttpOnly, Secure & SameSite Cookies
- SameSite=Lax vs Strict
- Why Tokens Must Not Live in localStorage
- BFF Cookie Sessions for SPAs

C9.3 — JWT Structure, Claims & Validation

7 nodes

JWT

TIER 1

- Header, Payload & Signature
- iss, aud, exp, sub, nbf & iat
- Signature Algorithms & alg:none
- JWKS & Signing-Key Rotation
- JWT vs Opaque Access Tokens
- Validating JWT Claims on Every Request

C9.4 — OAuth 2.0 Grants & Delegation

9 nodes

OAuth 2.0 Concepts

TIER 1

- Resource Owner, Client, Authorization Server & Resource Server
- Scopes & Least-Privilege Delegation
- Access Tokens
- Authorization Code Flow with PKCE
- Why Implicit Grant Is Deprecated
- Why Resource Owner Password Grant Is Deprecated
- Client Credentials for Service-to-Service
- Device Authorization Grant

C9.5 — OIDC Identity vs OAuth Authorization

6 nodes

OpenID Connect Concepts

TIER 1

- OAuth 2.0 Authorization vs OIDC Identity
- ID Tokens vs Access Tokens
- ID Token Claim Validation
- nonce Against Replay
- UserInfo Endpoint

C9.6 — Token Management, Rotation & Revocation

6 nodes

Token Management

TIER 1

- Short-Lived Access Tokens
- Refresh Token Rotation & Reuse Detection
- Token Revocation
- Bearer Header vs Cookie Transport
- Audience, Sender-Constrained Tokens & mTLS

C9.7 — Spring Security Filter Chain (Boot 3 / Security 6)

7 nodes

Spring Security

TIER 1

- SecurityFilterChain Beans
- OncePerRequestFilter
- SecurityContextHolder
- Security Context Across Async & Reactive Threads
- Multiple Filter Chains & Request Matchers

- OAuth2 Resource Server JWT

C9.8 — RBAC, ABAC & Method Security

7 nodes

RBAC

TIER 1

- Roles, Authorities & Scopes
- @PreAuthorize & SpEL
- Attribute-Based Access Control
- Domain-Level Ownership Checks
- Broken Access Control & IDOR
- @EnableMethodSecurity

C9.9 — XSS & Content-Security-Policy

6 nodes

XSS

TIER 1

- Reflected XSS
- Stored XSS
- DOM-based XSS
- Context-Aware Escaping & Framework Sanitization
- Content-Security-Policy

C9.10 — CSRF Mechanics

6 nodes

CSRF

TIER 1

- CSRF on Cookie-Authenticated Apps
- Why Bearer APIs Are CSRF-Immune
- Anti-CSRF Tokens
- SameSite as a CSRF Control
- Disabling CSRF Only for Stateless APIs

C9.11 — SQL Injection & Unsafe Persistence

5 nodes

SQL Injection

TIER 1

- Parameterized Queries
- Second-Order SQL Injection
- Unsafe Native Queries in Spring Data JPA
- String Concatenation in ORM Queries

C9.12 — CORS Misconfigurations

6 nodes

CORS (Security)

TIER 1

- CORS Is Browser Enforcement, Not a Firewall
- Preflight OPTIONS Requests
- Dynamic Origin Validation
- Access-Control-Allow-Credentials
- Wildcard Origin + Credentials Trap

C9.13 — Password Hashing

6 nodes

Password Hashing

TIER 1

- Argon2id
- Bcrypt Cost Factor ≥ 12
- Unique Per-User Salts
- Pepper vs Salt
- Spring PasswordEncoder

C9.14 — Secrets Management & Key Scanning

6 nodes

Secrets Management

TIER 1

- Runtime Ingest from Secrets Manager or Vault
- No Secrets in Git, Images or Logs
- CI/CD Scanning: GitGuardian & TruffleHog
- Secret Rotation

C9.15 — OWASP Architecture & Cryptographic Failures

6 nodes

OWASP Basics

TIER 1

- Broken Access Control (A01)
- Cryptographic Failures (A02)
- Defense in Depth: Gateway to Persistence
- Security Response Headers
- API Threat Modeling

C10 — Redis

16 sections

C10.1 — Redis Foundations & Why It Is Fast

7 nodes

Redis Foundations

TIER 1

- What Redis Is
- In-Memory Storage
- Single-Threaded Event Loop
- RESP Protocol
- Caching, Sessions & Real-Time Use Cases
- Real-World Application Patterns

C10.2 — Environment, Configuration & Clients

7 nodes

Environment & Configuration

TIER 1

- Install & Local Setup
- redis.conf Essentials
- Logical Databases (db 0...N)
- Client Connection Model
- Java / Lettuce / Jedis Clients
- Connection Pooling

C10.3 — Keys, Commands & Core Operations

5 nodes

Keys & Commands

TIER 1

- Key Naming Conventions
- EXISTS, DEL, TYPE, SCAN
- KEYS vs SCAN
- INFO, MONITOR & Server Introspection

C10.4 — Data Structures

9 nodes

Redis Data Structures

TIER 1

- Strings
- Hashes
- Lists
- Sets
- Sorted Sets
- HyperLogLog
- Bitmaps & Bitfields
- Choosing the Right Structure

C10.5 — Caching Patterns & Cache-Aside

6 nodes

Cache-Aside

TIER 1

- Cache Hit & Cache Miss
- Read-Through & Write-Through
- Write-Behind / Write-Back
- When Not to Cache

C10.6 — TTL & Expiration 5 nodes

TTL	TIER 1
— EXPIRE, PEXPIRE & TTL Commands	
— Volatile vs Persistent Keys	
— Passive vs Active Expiration	
— TTL Jitter to Avoid Synchronized Expiry	

C10.7 — Cache Invalidation 5 nodes

Cache Invalidation	TIER 1
— Invalidate-on-Write	
— Versioned / Namespaced Keys	
— Stale-While-Revalidate	
— Why Invalidation Is Hard	

C10.8 — Distributed Caching 5 nodes

Distributed Caching	TIER 1
— Local Cache vs Redis	
— Coherence Across Instances	
— Hot Keys & Skew	
— Maxmemory & Eviction Policies	

C10.9 — Sessions, Tokens & Feature State 5 nodes

Sessions & Token Storage	TIER 1
— Session Store Patterns	
— Storing Authentication Tokens	
— Feature Flags & Remote Config	
— Data Deduplication Keys	

C10.10 — Rate Limiting 5 nodes

Rate Limiting with Redis	TIER 1
— Fixed Window Counters	
— Sliding Window Patterns	
— Token Bucket / Leaky Bucket	
— Atomicity with INCR & Lua	

C10.11 — Distributed Locks & Inventory 5 nodes

Distributed Locks	TIER 2
— SET NX EX Lock Pattern	
— Redlock Trade-offs	
— Lock Safety, Fencing & TTL	
— Real-Time Inventory Reservation	

C10.12 — Pub/Sub, Queues & Real-Time Messaging 6 nodes

Pub/Sub	TIER 2
— PUBLISH / SUBSCRIBE	
— Pub/Sub Delivery Limits	
— Lists & Streams as Job Queues	
— Leaderboards with Sorted Sets	
— Redis Streams Overview	

C10.13 — Transactions, Pipelining & Scripting 5 nodes

Transactions, Pipelining & Scripting	TIER 2
--------------------------------------	--------

- MULTI / EXEC & WATCH
- Pipelining
- Lua Scripting
- Atomic Multi-Key Updates

C10.14 — Persistence & Backup

5 nodes

Redis Persistence

TIER 2

- RDB Snapshots
- AOF Append-Only File
- RDB vs AOF Trade-offs
- Backup & Restore

C10.15 — Replication, Partitioning & Clustering

5 nodes

Redis Replication

TIER 2

- Primary/Replica & Lag
- Partitioning Strategies
- Redis Cluster Basics
- Sentinel & Failover

C10.16 — Security, Memory & Operations

6 nodes

Security, Memory & Operations

TIER 2

- Auth, Network Binding & ACLs
- Benchmarks & Capacity Reality
- Memory & Fragmentation
- SLOWLOG & Latency Diagnosis
- Production Checklist

C11 — Kafka & Event-Driven Systems

16 sections

C11.1 — Kafka Foundations & Messaging Models

7 nodes

Kafka Foundations

TIER 1

- What Kafka Is
- Distributed Commit Log
- Kafka vs Traditional Messaging
- Point-to-Point vs Pub/Sub
- Analytics, Logs, Integration & Messaging Use Cases
- Events as Key-Value Records with Timestamps

C11.2 — Event-Driven Architecture

6 nodes

Event-Driven Architecture

TIER 1

- Events, Producers & Consumers
- Event Brokers & Loose Coupling
- Event Store & Replay
- CQRS with Event Streams
- Event Granularity & Best Practices

C11.3 — Cluster Architecture & Brokers

6 nodes

Cluster Architecture

TIER 1

- Brokers & Clusters
- Produce → Broker → Consume Workflow
- KRaft vs ZooKeeper Metadata
- Replication Factor & Leaders

- In-Sync Replicas (ISR)

C11.4 — Topics & Log Anatomy

6 nodes

Topics

TIER 1

- Topic Naming & Organization
- Log Anatomy: Append-Only Segments
- Time/Size Retention & Compaction
- Create & Describe Topics
- Log Compaction

C11.5 — Partitions & Parallelism

5 nodes

Partitions

TIER 1

- Key-Based Partitioning
- Leaders, Followers & Failover
- Choosing Partition Count
- Hot Partitions & Skew

C11.6 — Producers

7 nodes

Producers

TIER 1

- Producer API & Records
- acks=0 / 1 / all
- Batch Size & linger.ms
- Idempotent Producer
- Producer Retries & Delivery Timeout
- Key/Value Serialization

C11.7 — Consumers

6 nodes

Consumers

TIER 1

- Consumer API & poll()
- Subscribe vs Assign
- Consumer Lag
- Deserialization & Poison Messages
- max.poll.interval & Session Timeout

C11.8 — Consumer Groups & Rebalancing

6 nodes

Consumer Groups

TIER 1

- Group Load Balancing
- One Consumer per Partition per Group

Consumer Rebalancing

TIER 2

- Eager vs Cooperative Rebalancing
- Rebalance Storms & Sticky Assignment

C11.9 — Offsets & Retention Semantics

5 nodes

Offsets

TIER 1

- Auto vs Manual Commit
- commitSync vs commitAsync
- earliest / latest / specific Offset
- __consumer_offsets

C11.10 — Ordering Guarantees

5 nodes

Ordering Guarantees

TIER 1

- Order Within a Partition
- No Global Topic Order
- Key Affinity for Related Events

C11.11 — Delivery Semantics 5 nodes

At-Most/At-Least/Exactly-Once Concepts	TIER 1
— At-Least-Once	
— Exactly-Once Concepts	
— Kafka Transactions / EOS	
— Effectively-Once via Idempotent Consumers	

C11.12 — Idempotency, Retries & Dead-Letter Queues 6 nodes

Idempotent Consumers	TIER 1
— Deduplication Keys & Outbox Markers	
Retries	TIER 1
— Retry Topics & Backoff	
Dead-Letter Queues	TIER 1
— DLQ Inspection & Replay	

C11.13 — Schemas, Serialization & Evolution 5 nodes

Schema Evolution	TIER 1
— Avro, JSON & Protobuf	
— Schema Registry	
— Backward / Forward Compatibility	
— Avoiding Breaking Schema Changes	

C11.14 — Kafka Internals & Storage 5 nodes

Kafka Internals	TIER 2
— Segments, Rolling & Indexes	
— OS Page Cache & Zero-Copy	
— Controller & Metadata	
— MirrorMaker / Cross-Cluster Replication	

C11.15 — Connect, Streams & Ecosystem 6 nodes

Connect, Streams & Ecosystem	TIER 2
— Kafka Connect Source & Sink	
— Kafka Streams Basics	
— kSQL / Flink SQL Overview	
— Kafka vs Flink / Pulsar Boundaries	
— Java Client Libraries	

C11.16 — Outbox, Saga & Production Operations 7 nodes

Outbox Pattern	TIER 2
— Transactional Outbox with DB	
Saga Pattern	TIER 2
— Choreography vs Orchestration	
— Lag, Throughput & Alerting	
— TLS, SASL & ACLs	
— Production Checklist	

C12 — Backend Reliability 14 sections

C12.1 — Reliability Foundations & Failure Modes 7 nodes

Reliability Foundations	TIER 1
— Availability, Latency & Correctness	

- Partial Failure in Distributed Systems
- Blast Radius & Failure Domains
- Fail-Fast vs Fail-Safe
- Defense in Depth for Dependencies
- Critical vs Non-Critical Dependencies

C12.2 — Timeouts & Deadlines

6 nodes

Timeouts

TIER 1

- Connect vs Read/Write Timeouts
- Deadline / Timeout Propagation
- End-to-End Timeout Budgets
- Hanging Calls & Thread/Connection Exhaustion
- Choosing Timeout Defaults

C12.3 — Retries & Backoff

8 nodes

Retries

TIER 1

- Retryable vs Non-Retryable Errors
- Retry Amplification & Cascades
- Retry Budgets

Exponential Backoff

TIER 1

- Full / Equal / Decorrelated Jitter
- Max Attempts & Caps
- Retry-After & Server Hints

C12.4 — Idempotency

6 nodes

Idempotency

TIER 1

- Idempotency-Key Headers & Stores
- Naturally Idempotent Operations
- DB UNIQUE / Upsert Guards
- Effectively-Once Side Effects
- Idempotency Record TTL & Replay Windows

C12.5 — Rate Limiting & Throttling

6 nodes

Rate Limiting

TIER 1

- Client-Side vs Server-Side Limits
- Token Bucket & Leaky Bucket
- Concurrency Limits
- Load Shedding Under Overload
- Per-Tenant Fairness

C12.6 — Circuit Breakers

6 nodes

Circuit Breakers

TIER 1

- Closed, Open & Half-Open
- Error Rate & Slow-Call Thresholds
- Fallbacks When Open
- Breakers vs Retries Interaction
- Resilience4j / Spring Patterns

C12.7 — Bulkheads & Isolation

6 nodes

Bulkheads

TIER 1

- Thread-Pool / Semaphore Isolation
- Connection-Pool Isolation
- Queue & Consumer Isolation
- Noisy-Neighbor Containment

- Cell / Shard Isolation

C12.8 — Backpressure

5 nodes

Backpressure

TIER 1

- Bounded Queues & Rejection
- Slow Consumer Propagation
- Reactive / Async Backpressure Signals
- Unbounded Buffer Bloat

C12.9 — Graceful Degradation & Fallbacks

6 nodes

Graceful Degradation

TIER 1

- Disable Non-Critical Features
- Stale Cache / Last-Known-Good
- Safe Default Responses
- Priority Lanes Under Stress
- Honest UX During Degradation

C12.10 — Health Checks & Safe Rollouts

6 nodes

Health Checks

TIER 1

- Liveness vs Readiness
- Dependency Probes & Deep Checks
- Health-Check Anti-Patterns
- Graceful Shutdown & Drain
- Rolling Deploy Safety

C12.11 — Eventual Consistency & Compensation

6 nodes

Eventual Consistency

TIER 1

- Read-Your-Writes Expectations
- Compensating Actions
- Reconciliation Jobs
- User-Visible Consistency Trade-offs
- Inbox/Outbox for Reliable Side Effects

C12.12 — Failure Recovery

6 nodes

Failure Recovery

TIER 1

- Restart vs Self-Heal
- Poison Messages & Quarantine
- Runbooks & Manual Intervention
- Data Repair & Backfills
- Incident → Postmortem → Prevention

C12.13 — SLOs, Error Budgets & Reliability Signals

6 nodes

SLOs & Error Budgets

TIER 2

- SLI vs SLO vs SLA
- Error Budgets & Release Policy
- p50 / p95 / p99 for Reliability
- Saturation & Golden Signals
- Alert on User Symptoms

C12.14 — Chaos, Overload & Production Playbooks

6 nodes

Chaos, Overload & Playbooks

TIER 2

- Chaos / Fault Injection
- Game Days & Resilience Drills
- Overload Control Playbook
- Dependency Outage Playbook
- Production Reliability Checklist

C13.1 — Testing Strategy & the Pyramid

6 nodes

Testing Strategy

TIER 1

- Test Pyramid vs Ice-Cream Cone
- What to Test vs What Not to Test
- Risk-Based Test Selection
- Fast Feedback Loops
- Testing Trophy / Honeycomb Variants

C13.2 — Unit Testing

6 nodes

Unit Testing

TIER 1

- Arrange-Act-Assert
- Behavior vs Implementation Coupling
- Pure Domain / Pure Function Tests
- Parameterized & Property-Style Inputs
- Unit-Test Anti-Patterns

C13.3 — JUnit 5

6 nodes

JUnit

TIER 1

- Lifecycle: @BeforeEach / @AfterEach / Extensions
- Assertions & Assumptions
- @Nested & Display Names
- Extension Model
- JUnit 4 → 5 Migration Notes

C13.4 — Mockito & Test Doubles

6 nodes

Mockito

TIER 1

- Stubs, Mocks, Fakes & Spies
- verify() & Argument Captors
- When Not to Mock
- Strict Stubbing & Unnecessary Stubbing
- BDDMockito Style

C13.5 — Integration Testing

6 nodes

Integration Testing

TIER 1

- Narrow vs Broad Integration Tests
- Real Collaborators vs Doubles
- Test Data Setup & Cleanup
- Hermetic / Deterministic Integration
- Shared Fixtures & Pollution

C13.6 — API & Web-Layer Testing

6 nodes

API Testing

TIER 1

- MockMvc
- WebTestClient
- REST Assured / HTTP Client Tests
- Status, Payload, Headers & Errors
- AuthN/AuthZ in API Tests

C13.7 — Database Testing

6 nodes

Database Testing

TIER 1

- @DataJpaTest
- @Transactional Rollback Semantics
- Migration / Flyway Compatibility Checks

- Asserting SQL Side Effects
- H2 vs Real PostgreSQL Gaps

C13.8 — Testcontainers

6 nodes

Testcontainers

TIER 2

- GenericContainer & Modules
- PostgreSQL Container
- Reuse, Ryuk & CI Resource Cost
- Kafka / Redis Containers
- Testcontainers Anti-Patterns

C13.9 — Contract Testing

5 nodes

Contract Testing

TIER 2

- Consumer-Driven Contracts
- Pact / Spring Cloud Contract
- Schema Contracts (OpenAPI / AsyncAPI)
- Breaking Change Detection

C13.10 — Spring Boot Test Slices & Context

6 nodes

Spring Boot Testing

TIER 1

- @SpringBootTest
- @WebMvcTest
- @JsonTest
- @MockBean / @SpyBean Pitfalls
- ApplicationContext Caching & DirtiesContext

C13.11 — Async, Messaging & Time

6 nodes

Async & Messaging Tests

TIER 2

- Awaitility for Async Assertions
- Kafka Listener / Producer Tests
- Clock / Time Abstraction
- Flaky Test Diagnosis & Quarantine
- Concurrency & Race Tests

C13.12 — Load, Performance & Resilience Tests

6 nodes

Load Testing

TIER 2

- Latency SLOs in Load Tests
- k6 / JMeter / Gatling Overview
- Soak, Spike & Stress Profiles
- Performance Regression Gates
- Fault Injection in Test Envs

C13.13 — Coverage, Mutation & Quality Gates

5 nodes

Coverage & Quality Gates

TIER 2

- Line vs Branch Coverage Limits
- Mutation Testing (PIT)
- Static Analysis Alongside Tests
- Meaningful Assertions over Coverage Chasing

C13.14 — CI Strategy & Production Confidence

6 nodes

CI Testing Strategy

TIER 2

- Parallelization & Sharding
- PR Checks vs Nightly Suites
- Test Ownership & Failure Triage
- Shift-Left and Shift-Right Testing
- Production-Confidence Checklist

C14.1 — DevOps Foundations & Lifecycle

7 nodes

DevOps Foundations

TIER 1

- What Is DevOps
- DevOps Lifecycle
- DevOps vs Traditional SDLC
- Culture, Collaboration & Feedback Loops
- Automation, Workflow & Pipelines
- Evolution & Future Trends

C14.2 — Version Control for Delivery

5 nodes

Version Control for Delivery

TIER 1

- Version Control Systems
- Git Merge Strategies
- Trunk-Based vs GitFlow for CI
- Choosing a VCS

C14.3 — Linux Fundamentals for Containers

6 nodes

Linux Fundamentals

TIER 1

- Processes
- Files & Permissions
- Environment Variables
- Signals & Graceful Stop (SIGTERM)
- Shell Debugging on Hosts & Containers

C14.4 — Containers & Docker Architecture

7 nodes

Containers

TIER 1

- What Is Docker
- Containers vs Virtual Machines
- Docker Architecture (Client, Daemon, Images)
- Essential Docker Commands
- How Docker Fits DevOps & CI/CD
- Common Docker Challenges in Teams

C14.5 — Docker Images & Registries

6 nodes

Docker Images

TIER 1

- Image Layers & Build Cache
- Docker Hub & Public Repositories
- Private Registries (ECR / Harbor)
- Push, Pull & Semantic Tagging
- Digests vs :latest in Production

C14.6 — Dockerfile & Image Optimization

8 nodes

Dockerfile

TIER 1

- Dockerfile Syntax
- .dockerignore & Build Context
- Image Size & Layer Best Practices
- Non-Root USER & Secure Defaults

Multi-stage Builds

TIER 1

- Builder Stage vs Runtime Stage

C14.7 — Docker Compose & Local Stacks 5 nodes

Docker Compose	TIER 1
— Services, Networks & Volumes in YAML	
— depends_on & Healthchecks	
— Scaling Services Locally	
— Why Compose Is Not Production Orchestration	

C14.8 — Docker Networking 5 nodes

Docker Networking	TIER 1
— Bridge Network (Default & User-Defined)	
— Host, Overlay & Macvlan Modes	
— Port Publishing & EXPOSE	
— Service DNS Between Containers	

C14.9 — Volumes & Container Storage 5 nodes

Volumes	TIER 1
— Named Volumes vs Bind Mounts	
— Sharing Data Between Containers	
— Storage Drivers & Data Persistence	
— Backup & Volume Lifecycle	

C14.10 — CI/CD Concepts & Deployment Automation 6 nodes

CI/CD Concepts	TIER 1
— Continuous Integration vs Delivery vs Deployment	
— Deployment Automation	
— Immutable Artifacts (Image by Digest)	
— Build → Test → Scan → Push → Deploy	
— Docker as the Build & Test Environment	

C14.11 — Pipeline Tools: GitHub Actions & Jenkins 5 nodes

GitHub Actions / Jenkins	TIER 2
— GitHub Actions Workflow Basics	
— Jenkins Pipeline Overview	
— CI Secrets & OIDC to Cloud	
— Dependency & Layer Caching in CI	

C14.12 — Kubernetes Orchestration 7 nodes

Kubernetes	TIER 2
— Why Container Orchestration	
— Pods, Deployments, Services & Ingress	
— Docker vs Kubernetes Responsibilities	
— Rolling Updates, Probes & Self-Healing	
— DevOps Best Practices on Kubernetes	
— Docker Swarm (Legacy Contrast)	

C14.13 — AWS ECS & Managed Containers 4 nodes

AWS ECS for Container Workloads	TIER 2
— Tasks, Services & Task Definitions	
— ECS/Fargate vs EKS Trade-offs	
— Build → ECR → ECS Deploy Flow	

C14.14 — Infrastructure as Code 5 nodes

Infrastructure as Code	TIER 2
------------------------	--------

- Why IaC Beats Click-Ops
- Introduction to Terraform
- AWS CloudFormation
- State, Drift & Reviewable Infra Changes

C14.15 — Monitoring & Logging for Delivery

5 nodes

Monitoring & Logging for Delivery

TIER 2

- Prometheus & Grafana with Helm
- Monitoring & Logging Services Overview
- Golden Signals in Deploy Pipelines
- Incident Comms (ChatOps Patterns)

C14.16 — DevSecOps & Container Security

6 nodes

DevSecOps & Container Security

TIER 2

- What Is DevSecOps
- Image CVE Scanning & SBOM
- Docker Security Best Practices
- Kubernetes Security Basics
- Secrets in Images, Compose & Pipelines

C15 — AWS / Cloud

16 sections

C15.1 — Cloud Foundations & Global Infrastructure

7 nodes

Cloud Foundations & AWS

TIER 1

- What Is Cloud Computing
- What Is AWS
- Regions, Availability Zones & Edge Locations
- Shared Responsibility Model
- Management Console & AWS CLI
- Free Tier & Account Setup Basics

C15.2 — IAM & Account Security

7 nodes

IAM

TIER 1

- Users, Groups & Roles
- Policies & Least Privilege
- Instance Profiles & Roles for EC2
- MFA for Root & Privileged Users
- Federation & SAML Overview
- Explicit Deny & Guardrail Policies

C15.3 — Secrets Manager & Credential Hygiene

5 nodes

Secrets Manager

TIER 2

- Runtime Secret Injection
- Secret Rotation
- Secrets Manager vs Parameter Store
- No Secrets in AMIs, Images or User Data

C15.4 — EC2 Compute

8 nodes

EC2

TIER 1

- Launching & Managing Instances
- Instance Types & Families
- On-Demand, Reserved & Spot Pricing
- AMIs & Golden Images
- Connecting to Linux/Windows Instances
- User Data & IMDSv2

C15.5 — Lambda & Serverless	6 nodes
Lambda	TIER 2
— Event Sources (S3, API Gateway, SQS)	
— Creating & Deploying Functions	
— Timeouts, Memory & Cold Starts	
— Execution Roles & Least Privilege	
— API Gateway Basics	
C15.6 — Containers: ECS, EKS & ECR	7 nodes
ECS	TIER 2
— Tasks, Services & Task Definitions	
— ECR Image Registry	
— Fargate vs EC2 Launch Type	
EKS	TIER 2
— EKS vs ECS Trade-offs	
— Managed Control Plane Basics	
C15.7 — S3 Object Storage	6 nodes
S3	TIER 1
— Buckets, Objects & Keys	
— Storage Classes	
— Lifecycle Rules	
— Bucket Policies, Block Public Access & Encryption	
— Static Website & Asset Hosting	
C15.8 — EBS, EFS & Storage Comparison	6 nodes
EBS, EFS & Storage Choices	TIER 1
— EBS Volumes, Attach & Snapshots	
— EFS Shared File Storage	
— EBS vs EFS vs S3	
— Glacier & Backup / DR Overview	
— Storage Gateway & Snow Family Overview	
C15.9 — VPC Networking & Security	6 nodes
VPC Fundamentals	TIER 1
— Subnets, Route Tables & Internet/NAT Gateways	
— Security Groups	
— NACLs vs Security Groups	
— VPC Peering & Gateway/Interface Endpoints	
— Bastion Hosts & Private Access Patterns	
C15.10 — Load Balancing & Auto Scaling	7 nodes
Load Balancers (AWS)	TIER 1
— ALB vs NLB vs Classic	
— Target Groups & Health Checks	
— Listeners, TLS & Path Routing	
Auto Scaling	TIER 1
— ASG Scaling Policies	
— Health-Based Instance Replacement	
C15.11 — Route 53 & CloudFront	6 nodes
Route 53	TIER 2
— Records & Routing Policies	

— Health Checks & DNS Failover	
CloudFront	TIER 2
— CDN Caching & Origins	
— HTTPS, OAI/OAC & Static Assets	

C15.12 — RDS & Aurora	6 nodes
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RDS	TIER 1
— Engines, Create & Connectivity	
— Multi-AZ & Read Replicas	
— Automated Backups & Snapshots	
— Aurora Overview	
— RDS vs Aurora	

C15.13 — DynamoDB & Managed Data Stores	6 nodes
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DynamoDB	TIER 2
— Tables, Partition & Sort Keys	
— On-Demand vs Provisioned Capacity	
— Streams & Lambda Integration	
— ElastiCache Overview	
— Redshift Overview	

C15.14 — SQS, SNS & Application Integration	7 nodes
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SQS	TIER 2
— Standard vs FIFO Queues	
— Visibility Timeout & DLQ	
SNS	TIER 2
— Topics & Subscriptions	
— SNS → SQS Fan-out Pattern	
— EventBridge & Kinesis Overview	

C15.15 — CloudWatch, CloudTrail & Operations	6 nodes
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CloudWatch	TIER 1
— Metrics, Alarms & Dashboards	
— CloudWatch Logs	
— CloudTrail vs CloudWatch	
— Synthetics & Canaries	
— Trusted Advisor & AWS Health Overview	

C15.16 — Cost, IaC & Well-Architected	6 nodes
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Cost, IaC & Well-Architected	TIER 2
— Cost Explorer, Budgets & Billing Alarms	
— CloudFormation Overview	
— Well-Architected Framework Pillars	
— Organizations & Account Governance Basics	
— Migration Strategies Overview	

C16 — Observability	14 sections
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C16.1 — Observability Foundations	6 nodes
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Observability Foundations	TIER 1
— Logs, Metrics & Traces	
— Monitoring vs Observability	
— USE, RED & Golden Signals	
— Emit → Collect → Store → Query → Act	

C16.2 — Structured Logging	6 nodes
Structured Logging	TIER 1
— JSON Log Schema & Stable Fields	
— MDC / Request Context	
— Levels, Sampling & Volume Control	
— PII Redaction & Secrets in Logs	
— Aggregation: Loki / ELK / CloudWatch Logs	
C16.3 — Metrics Fundamentals	6 nodes
Metrics	TIER 1
— Counters & Gauges	
— Histograms, Summaries & Buckets	
— Labels / Dimensions	
— Pull vs Push Metrics	
— Micrometer / Spring Metrics	
C16.4 — Golden Signals & SLIs	9 nodes
Latency Percentiles	TIER 1
— p50 / p95 / p99 Tail Latency	
— Why Averages Lie	
Error Rates	TIER 1
— Availability as an SLI	
Throughput	TIER 1
— RPS, Concurrency & Saturation Link	
Health Metrics	TIER 1
— CPU, Memory, Queue & Pool Saturation	
C16.5 — Distributed Tracing	6 nodes
Distributed Tracing	TIER 1
— Traces, Spans & Parent-Child Links	
— Attributes, Events & Status	
— Head vs Tail Sampling	
— Jaeger / Zipkin / Tempo / X-Ray	
— Trace-Driven Debugging Across Services	
C16.6 — Correlation & Context Propagation	5 nodes
Correlation IDs	TIER 1
— traceparent, W3C Trace Context & Baggage	
— HTTP / Messaging Header Propagation	
— Joining Logs ↔ Metrics ↔ Traces	
— Async / Thread-Pool Context Loss	
C16.7 — OpenTelemetry	6 nodes
OpenTelemetry Concepts	TIER 1
— OTel Signals: Logs, Metrics, Traces	
— SDK vs Auto-Instrumentation	
— OpenTelemetry Collector	
— Exporters & Vendors	
— Semantic Conventions	
C16.8 — Prometheus	6 nodes
Prometheus	TIER 2
— PromQL Basics	

- Scraping & Service Discovery
- Recording Rules
- Prometheus Cardinality Explosions
- HA, Federation & Long-Term Storage

C16.9 — Grafana, Dashboards & Visualization

5 nodes

Grafana

TIER 2

- Dashboard Design for Incidents
- RED/USE Dashboard Patterns
- Variables & Multi-Service Boards
- Dashboards as Code

C16.10 — Alerting & On-Call

6 nodes

Alerting & On-Call

TIER 2

- Alert on Symptoms, Not Causes
- Alert Noise, Fatigue & Routing
- Multi-Window Burn-Rate Alerts
- Runbooks Linked from Alerts
- Pager Hygiene & Escalation

C16.11 — Frontend, RUM & Full-Stack Telemetry

6 nodes

Frontend & RUM Observability

TIER 1

- Core Web Vitals (LCP, INP, CLS)
- Real User Monitoring
- Frontend Error Tracking
- Linking Browser Spans to Backend Traces
- API Latency from the Client Perspective

C16.12 — Cardinality, Cost & Sampling

5 nodes

Cardinality, Cost & Sampling

TIER 2

- Dangerous High-Cardinality Labels
- Log Volume & Retention Cost
- Adaptive / Tail-Based Sampling
- Telemetry Budgets per Service

C16.13 — Profiling & Continuous Diagnostics

5 nodes

Profiling & Continuous Diagnostics

TIER 2

- CPU & Heap Profiles
- Continuous Profiling
- JVM Runtime Signals (GC, Threads)
- Catching Perf Regressions in Prod

C16.14 — Incident Debugging Playbooks

6 nodes

Incident Debugging Playbooks

TIER 2

- Symptom → Signal → Hypothesis → Prove
- Cross-Service Investigation
- Customer-Impact-First Triage
- Post-Incident Telemetry Gaps
- Observability Interview Checklist

LLD, System Design & Distributed Systems

OOD, patterns, LLD, HLD, distributed systems. 80 sections · 456 study nodes.

D1 — Object-Oriented Design

7 nodes

Encapsulation	TIER 1
Abstraction	TIER 1
Composition vs Inheritance	TIER 1
SOLID	TIER 1
Coupling/Cohesion	TIER 1
Interfaces (Design)	TIER 1
Immutability (Design)	TIER 1

D2 — Design Patterns

13 nodes

Strategy	TIER 1
Factory	TIER 1
Builder	TIER 1
Observer	TIER 1
Adapter	TIER 1
Decorator	TIER 1
Facade	TIER 1
Command	TIER 2
Template Method	TIER 2
Chain of Responsibility	TIER 2
State	TIER 2
Proxy	TIER 2
Visitor	TIER 3

D3 — LLD Interview Problems

14 nodes

Parking Lot	TIER 1
Elevator	TIER 1
Library	TIER 1
Vending Machine	TIER 1
Tic-Tac-Toe	TIER 1
Chess	TIER 1
Splitwise	TIER 1
ATM	TIER 1
Notification Service (LLD)	TIER 1
Logging Framework	TIER 1
Rate Limiter (LLD)	TIER 1
Cache (LLD)	TIER 1
Movie Ticket Booking (LLD)	TIER 2
Food Delivery (LLD)	TIER 2

D4 — Distributed Systems Foundations

12 sections

D4.1 — System Design Foundations, HLD & LLD Boundaries

6 nodes

System Design Foundations

TIER 1

- What System Design Is
- High-Level vs Low-Level Design
- Components, Interfaces & Data Flow
- Scalability, Reliability, Performance & Maintainability
- Design for Requirements, Not Buzzwords

D4.2 — Requirements, Constraints & Trade-offs

6 nodes

Requirements & Constraints

TIER 1

- Functional Requirements
- Non-Functional Requirements & SLOs
- Read/Write Mix, Access Patterns & Data Shape
- Latency, Freshness, Availability & Cost Trade-offs
- Scope, Assumptions & Explicit Non-Goals

D4.3 — Scale from One Server to Millions

7 nodes

Scale from One Server to Millions

TIER 1

- Single-Server Baseline
- Separate Web & Data Tiers
- Horizontal vs Vertical Scaling
- Stateless vs Stateful Services
- Redundancy at Every Tier
- Multi-Datacenter Evolution

D4.4 — Latency, Throughput & Performance Budgets

6 nodes

Latency & Throughput

TIER 1

- p50, p95, p99 & Tail Latency
- Throughput, Concurrency & Utilization
- Queueing, Saturation & Little's Law
- Memory vs Latency
- Batching, Compression & Fewer Round Trips

D4.5 — Load Balancing, Proxies & Traffic Routing

7 nodes

Load Balancing

TIER 1

- Layer 4 vs Layer 7 Load Balancing
- Round Robin, Least Connections & Weighted Routing
- Health Checks & Failure Removal
- Sticky Sessions & Why Stateless Is Easier
- Forward vs Reverse Proxy
- GeoDNS, Anycast & Global Routing

D4.6 — Availability, Reliability & Failure Models

7 nodes

Availability

TIER 1

- Reliability vs Availability
- Crash, Omission, Timing & Byzantine Failures
- Partial Failure Is the Default
- Single Points of Failure
- Fault Tolerance, Redundancy & Graceful Degradation
- Availability Math & Dependency Composition

D4.7 — CAP, PACELC & Consistency Models

8 nodes

CAP Theorem

TIER 1

- Consistency
- Strong vs Eventual Consistency
- Eventual Consistency
- PACELC: Partition and Normal-Operation Trade-offs
- Read-Your-Writes, Monotonic Reads & Session Guarantees
- Causal Consistency
- Linearizability vs Serializability

D4.8 — Replication, Leaders & Quorums

7 nodes

Replication

TIER 1

- Leader/Follower Architecture
- Synchronous vs Asynchronous Replication
- Single-Leader vs Multi-Leader
- Read/Write Quorums
- Replication Lag & Stale Reads
- Conflict Detection & Resolution

D4.9 — Partitioning, Sharding & Consistent Hashing

8 nodes

Partitioning & Sharding

TIER 1

- Range vs Hash Partitioning
- Choosing a Shard / Partition Key
- Hot Keys, Celebrity Problem & Skew
- Resharding & Online Data Movement
- Consistent Hashing
- Virtual Nodes & Bounded Loads
- Cross-Shard Queries, Joins & Transactions

D4.10 — Distributed Time, Ordering & Consensus

8 nodes

Consensus Concepts

TIER 1

- Distributed Clock & Time Issues
- Lamport & Vector Clocks
- Total vs Partial Ordering
- Leader Election
- Gossip Protocols
- Raft Details
- Split Brain, Leases & Fencing Tokens

D4.11 — Delivery Semantics & Idempotency

6 nodes

Idempotency in Distributed Systems

TIER 1

- At-Most, At-Least & Effectively-Once Delivery
- Deduplication Keys & Inbox Tables
- Retries and Non-Idempotent Side Effects
- Ordering, Duplicates & Consumer Correctness
- Why “Exactly Once” Has a Scope

D4.12 — Distributed Transactions & Workflows

6 nodes

Distributed Transactions

TIER 1

- Two-Phase Commit
- Saga: Orchestration vs Choreography
- Transactional Outbox & CDC
- Compensation & Semantic Rollback
- Durable Workflow Engines

D5 — Data & Storage Architecture

10 sections

D5.1 — Data Modeling & Access-Pattern-First Design

7 nodes

Data Modeling for Systems	TIER 1
— Entities, Relationships & Invariants	
— Model from Access Patterns	
— Normalization vs Denormalization	
— Secondary Indexes & Write Amplification	
— Retention, Archival, Deletion & Compliance	
— Data Retention Policies	
D5.2 — SQL, NoSQL & Database Selection	7 nodes
SQL vs NoSQL	TIER 1
— Relational Databases	
— Key-Value Stores	
— Document Databases	
— Wide-Column Stores	
— Graph Databases	
— Polyglot Persistence & Its Cost	
D5.3 — Database Replication & Read Scaling	5 nodes
Database Replication	TIER 1
— Read Replicas	
— Read-After-Write Routing	
— Failover, Promotion & Data Loss Windows	
— Cross-Region Replication	
D5.4 — Database Partitioning & Sharding	6 nodes
Database Sharding	TIER 1
— Data Partitioning	
— Hot Partitions	
— Shard Routing & Metadata	
— Global Secondary Indexes	
— Online Schema & Shard Migrations	
D5.5 — Block, File & Object Storage	6 nodes
Blob & Object Storage	TIER 1
— Block vs File vs Object Storage	
— Objects, Metadata & Immutable Blobs	
— Pre-Signed Upload / Download Flows	
— Multipart Uploads, Chunking & Checksums	
— Durability, Replication & Erasure Coding	
D5.6 — Caching Architecture & Invalidation	8 nodes
Caching	TIER 1
— Local vs Distributed Cache	
— Cache-Aside, Read-Through & Write-Through	
— Write-Back / Write-Behind	
— LRU, LFU, FIFO & TTL	
— Invalidation, Versioned Keys & Freshness	
— Stampede, Penetration & Hot Keys	
— Cold Starts, Warming & Precomputation	
D5.7 — CDN & Edge Delivery	6 nodes
Content Delivery Networks	TIER 1
— Edge PoPs, Origins & Cache Keys	
— TTL, Invalidation & Versioned Assets	
— Static vs Dynamic Content Caching	
— Origin Shielding & CDN Failure Fallback	

- Edge Compute & Personalization

D5.8 — Queues, Pub/Sub & Async Work

7 nodes

Message Queues

TIER 1

- Queue vs Publish/Subscribe
- Producer, Broker & Consumer
- Load Leveling & Backpressure Buffer
- Acknowledgements, Visibility Timeout & DLQ
- Ordering, Partitions & Consumer Groups
- Poison Messages, Retry & Replay

D5.9 — Event Streams & Data Pipelines

7 nodes

Event Streams

TIER 1

- Append-Only Log vs Queue
- Event Streaming & Consumer Offsets
- Batch vs Stream Processing
- Event Sourcing vs Event Streaming
- CDC & Materialized Views
- Lambda vs Kappa Architecture

D5.10 — Search, Indexes & Probabilistic Structures

7 nodes

Search Systems

TIER 1

- Inverted Index
- Tries & Prefix Search
- Retrieval, Ranking & Relevance
- Bloom Filters
- Indexing Pipeline & Near-Real-Time Refresh
- Search Sharding & Replication

D6 — API & Communication Design

6 sections

D6.1 — API Contracts & Protocol Selection

6 nodes

API Design Foundations

TIER 1

- REST
- gRPC
- GraphQL Concepts
- REST vs gRPC vs GraphQL
- Synchronous vs Asynchronous APIs

D6.2 — Resource Modeling, Errors & Compatibility

6 nodes

Resource Modeling

TIER 1

- API Versioning & Evolution
- Error Modeling
- Backward & Forward Compatibility
- Schema Contracts & Validation
- Partial Success & Bulk APIs

D6.3 — Pagination, Filtering & Large Results

6 nodes

Pagination

TIER 1

- Offset vs Cursor / Keyset Pagination
- Filtering & Sorting
- Stable Sort Keys
- Pagination under Concurrent Writes

- Async Exports for Huge Result Sets

D6.4 — Real-Time Communication

5 nodes

WebSockets

TIER 2

- Server-Sent Events
- Polling vs Long Polling
- Connection Gateways, Fan-out & Presence
- Reconnect, Resume Tokens & Missed Events

D6.5 — API Gateway, Security & Rate Limiting

6 nodes

API Gateway & Edge Controls

TIER 1

- Authentication & Authorization
- Rate Limiting
- Idempotency Keys
- API Gateway vs Load Balancer
- Validation, Quotas & WAF

D6.6 — Service Communication & Discovery

6 nodes

Service Communication

TIER 1

- Service Discovery & Registration
- Client-Side vs Server-Side Discovery
- Heartbeats, Leases & Health
- Service Mesh & Sidecars
- Consumer-Driven Contract Testing

D7 — Scalability, Reliability & Architecture

8 sections

D7.1 — Architecture Styles & Service Boundaries

8 nodes

Architecture Styles

TIER 1

- Monolith & Modular Monolith
- Microservices
- Monolith vs Microservices Trade-offs
- Event-Driven Architecture
- Serverless Architecture
- Service Boundaries & Conway's Law
- Strangler Migration from Monolith

D7.2 — Caching Strategy at System Scale

5 nodes

Caching Strategies

TIER 1

- Cache Invalidation
- Browser, CDN, Service & Database Caches
- Cache Consistency & Write Policies
- Cache Failure, Stampede & Bypass

D7.3 — Overload Protection & Backpressure

6 nodes

Overload Control

TIER 1

- Distributed Rate Limiting
- Load Shedding
- Backpressure
- Bounded Queues & Admission Control
- Priority, Fairness & Tenant Isolation

D7.4 — Timeouts, Retries & Circuit Breakers

7 nodes

Failure Containment

TIER 1

- Timeouts & Deadline Propagation
- Retries, Backoff & Jitter
- Circuit Breakers
- Bulkheads & Failure Domains
- Cascading Failure Analysis
- Fallbacks & Graceful Degradation

D7.5 — High Availability, Failover & Disaster Recovery

7 nodes

High Availability

TIER 1

- Single Points of Failure
- Redundancy
- Failover
- Disaster Recovery
- RTO, RPO, Backups & Restore Testing
- Active/Active vs Active/Passive

D7.6 — SLIs, SLOs & Production Readiness

7 nodes

SLI, SLO & SLA Concepts

TIER 1

- Service-Level Objectives
- Service-Level Agreements
- Availability Calculations
- Error Budgets & Release Policy
- Latency, Traffic, Errors & Saturation
- Production Readiness Review & Runbooks

D7.7 — Multi-Region Architecture

6 nodes

Multi-Region Architecture

TIER 2

- Global Traffic Routing & Locality
- Data Residency & Sovereignty
- Cross-Region Replication & Conflict Handling
- Region Evacuation & Failover Drills
- Cell-Based Architecture

D7.8 — Security, Privacy, Cost & Sustainability

7 nodes

Cross-Cutting Architecture Concerns

TIER 2

- Threat Modeling & Trust Boundaries
- Zero Trust & Least Privilege
- Encryption & Key Management
- Privacy, Retention & Right to Delete
- Cost vs Performance Optimization
- Sustainability & Resource Efficiency

D8 — Capacity Estimation

4 sections

D8.1 — Back-of-the-Envelope Estimation Method

5 nodes

Back-of-the-Envelope Estimation

TIER 1

- Powers of Two, Time & Latency Numbers
- Average vs Peak & Safety Margins
- Assumption-Driven Estimates
- Order-of-Magnitude over False Precision

D8.2 — Traffic, QPS & Concurrency Estimation

5 nodes

Requests per Second

TIER 1

- Read/Write Ratios
- DAU → Requests/Day → Average/Peak QPS

- Concurrent Connections
- Fan-out & Internal Request Amplification

D8.3 — Storage, Bandwidth & Cache Estimation

6 nodes

Storage Estimation

TIER 1

- Bandwidth & Egress Estimation
- Cache-Size Estimation
- Database-Growth Estimation
- Replication, Index & Metadata Overhead
- Hot, Warm & Cold Retention Tiers

D8.4 — Capacity Planning & Bottleneck Analysis

5 nodes

Server & Capacity Requirements

TIER 2

- CPU, Memory, Disk I/O & Network Limits
- Utilization Targets & Headroom
- Capacity Model & Load-Test Validation
- Cost Envelope & Unit Economics

D9 — System Design Interview Method

7 sections

D9.1 — The 45–60 Minute Interview Framework

6 nodes

System Design Interview Framework

TIER 1

- Time Budget by Interview Phase
- Clarify Before Drawing Boxes
- Top-Down Design & Iterative Refinement
- Drive the Interview Collaboratively
- Maintain Assumptions & Decision Log

D9.2 — Requirements & Scope in Interviews

6 nodes

Requirements Gathering

TIER 1

- Functional Requirements
- Non-Functional Requirements
- Prioritize Core Use Cases
- Scale, Geography & Tenant Model
- Consistency, Latency & Availability Targets

D9.3 — Estimation, APIs & Data Model in Interviews

6 nodes

Capacity Estimation in Interviews

TIER 1

- API Design
- Data Modeling
- Estimate What Changes the Design
- Idempotency, Pagination & Error Contracts
- Schema from Critical Access Patterns

D9.4 — High-Level Architecture & Request Flows

6 nodes

High-Level Architecture

TIER 1

- Critical Read Path
- Critical Write Path
- Synchronous vs Asynchronous Boundaries
- Component Responsibilities & Ownership
- Diagram Legibility & Data-Flow Labels

D9.5 — Deep Dives, Scaling & Failure Handling

7 nodes

Component Deep Dives

TIER 1

- Scaling the Bottleneck
- Reliability & Failure Handling
- What If This Node or Region Dies?
- Consistency & Correctness
- Security & Privacy
- Observability & Operations

D9.6 — Trade-offs & Senior-Level Communication

6 nodes

Trade-off Analysis

TIER 1

- Compare Viable Alternatives
- Name Residual Risk & Revisit Triggers
- Avoid Premature Microservices & Sharding
- Defend Choices and Adapt to Pushback
- Ownership, Migration & Operational Cost

D9.7 — Mocks, Deliberate Practice & Company Calibration

7 nodes

System Design Deliberate Practice

TIER 2

- Whiteboard / Excalidraw Fluency
- Timed Mock Interviews & Feedback
- Design Retrospective & Gap Log
- Compare Multiple Solutions to One Problem
- Company/Level-Specific Calibration
- Build, Scale, Measure, Break & Rebuild

D10 — System Design Practice

30 sections

D10.1 — Design a URL Shortener

4 nodes

URL Shortener

TIER 1

- ID Generation & Base62
- Read-Heavy Cache & Redirect Path
- Expiry, Analytics & Abuse Prevention

D10.2 — Design Pastebin

4 nodes

Pastebin

TIER 1

- Metadata DB + Object Storage
- Expiration & Cleanup
- Hot Content & CDN

D10.3 — Design a Distributed Rate Limiter

4 nodes

Distributed Rate Limiter

TIER 1

- Token Bucket, Leaky Bucket & Sliding Window
- Distributed Counters & Atomicity
- Fail-Open vs Fail-Closed

D10.4 — Design a Key-Value Store

4 nodes

Distributed Key-Value Store

TIER 1

- Partitioning & Replication
- Consistency, Quorums & Conflict Resolution
- WAL, Memtable & SSTable Concepts

D10.5 — Design a Unique ID Generator

4 nodes

Distributed Unique ID Generator

TIER 1

- UUID vs Database Sequence vs Snowflake
- Timestamp, Node & Sequence Bits

- Clock Rollback & Worker-ID Coordination

D10.6 — Design a Notification System	4 nodes
Notification System — Email, SMS, Push & In-App Channels — Priority, Retry, DLQ & Deduplication — Preferences, Templates & Rate Limits	TIER 1
D10.7 — Design a News Feed	4 nodes
Twitter / News Feed — Fan-out on Write vs Read — Celebrity / Hot-Key Problem — Ranking, Cache & Cursor Pagination	TIER 1
D10.8 — Design a Chat System	4 nodes
Chat System — WebSocket Gateways & Session Routing — Message Ordering, Delivery & Read Receipts — Presence, Offline Delivery & Multi-Device Sync	TIER 1
D10.9 — Design Google Drive / Dropbox	4 nodes
Dropbox / Google Drive — Chunking, Deduplication & Multipart Upload — Change Sync & Conflict Resolution — Metadata DB vs Blob Storage	TIER 1
D10.10 — Design YouTube / Video Streaming	4 nodes
YouTube / Video Streaming — Upload, Transcode & Processing Pipeline — CDN & Adaptive Bitrate Streaming — Metadata, Views & Recommendation Boundaries	TIER 1
D10.11 — Design Search Autocomplete	4 nodes
Search Autocomplete — Trie / Prefix Index & Top-K — Ranking, Freshness & Personalization — Caching, Sharding & Hot Prefixes	TIER 1
D10.12 — Design a Ticket Booking System	4 nodes
Ticket Booking System — Seat Holds, Locks & Expiration — Prevent Overselling under Concurrency — Flash-Sale Queues, Fairness & Idempotent Payment	TIER 1
D10.13 — Design Uber / Ride Hailing	4 nodes
Uber / Ride Hailing — Location Updates & Geospatial Indexing — Driver Matching & Dispatch — Trip State Machine, Pricing & Event Flow	TIER 1
D10.14 — Design a Payment System	4 nodes
Payment System — Double-Entry Ledger & Immutable Events	TIER 1

- Idempotency & Exactly-Once Business Effect
- PSP Workflow, Reconciliation & Refunds

D10.15 — Design a Distributed Job Scheduler

4 nodes

Distributed Job Scheduler

TIER 1

- Scheduling Index / Timing Wheel
- Leases, Ownership & Failover
- Retry, Deduplication & Misfires

D10.16 — Design a Monitoring & Alerting Platform

4 nodes

Metrics Monitoring & Alerting Platform

TIER 1

- High-Throughput Metric Ingestion
- Time-Series Storage & Downsampling
- Alert Evaluation, Routing & Deduplication

D10.17 — Design a Distributed Cache

4 nodes

Distributed Cache

TIER 1

- Partition, Replicate & Rebalance
- Eviction, Hot Keys & Stampede
- Consistency & Node Failure

D10.18 — Design an Ad-Click Aggregation System

4 nodes

Ad-Click Event Aggregation

TIER 2

- Event Ingestion & Partitioning
- Windowed Aggregation & Late Events
- Deduplication & Batch Reconciliation

D10.19 — Design a Web Crawler

4 nodes

Web Crawler

TIER 2

- URL Frontier, Priority & Politeness
- URL/Content Deduplication & Bloom Filters
- Distributed Fetchers, Retry & Storage

D10.20 — Design a Proximity / Nearby Service

4 nodes

Proximity / Nearby Service

TIER 2

- Geohash, Quadtree & S2 Cells
- Radius Search & Boundary Expansion
- Location Updates, Hotspots & Caching

D10.21 — Design Google Maps

4 nodes

Maps & Navigation

TIER 2

- Map Tiles, Object Storage & CDN
- Road Graph, Routing & Precomputation
- Live Traffic & ETA Updates

D10.22 — Design a Distributed Message Queue

4 nodes

Distributed Message Queue

TIER 2

- Partitions, Ordering & Replication
- Offsets, Acknowledgements & Consumer Groups
- Retention, Replay & Rebalancing

D10.23 — Design a Distributed Email Service

4 nodes

Distributed Email Service

TIER 2

- Queues, Priority & Provider Routing
- Rate Limits, Reputation & Bounce Handling
- Templates, Tracking & Idempotency

D10.24 — Design S3-Like Object Storage

4 nodes

S3-Like Object Storage

TIER 2

- Metadata Plane vs Data Plane
- Chunk Placement, Replication & Erasure Coding
- Consistency, Checksums & Repair

D10.25 — Design a Real-Time Leaderboard

4 nodes

Real-Time Gaming Leaderboard

TIER 2

- Sorted Sets & Rank Queries
- Sharding, Top-K & Global Merge
- Seasons, Ties & Anti-Cheat

D10.26 — Design a Digital Wallet

4 nodes

Digital Wallet

TIER 2

- Ledger vs Derived Balance
- Transfers, Idempotency & Saga
- Audit, Reconciliation & Fraud Controls

D10.27 — Design a Stock Exchange

4 nodes

Stock Exchange

TIER 3

- Order Book & Price-Time Priority
- Deterministic Matching & Sequencing
- Market Data Fan-out, Durability & Audit

D10.28 — Design an Online Coding Judge

4 nodes

Online Coding Judge

TIER 2

- Secure Sandboxing & Resource Isolation
- Submission Queue & Language Workers
- Test Storage, Scoring & Result Streaming

D10.29 — Design Collaborative Documents

4 nodes

Google Docs / Collaborative Editing

TIER 3

- Operational Transform vs CRDT
- Presence, Offline Edits & Synchronization
- Snapshots, History & Compaction

D10.30 — Design an E-Commerce / Food Delivery Platform

5 nodes

E-Commerce & Food Delivery Platform

TIER 2

- Catalog, Search & Inventory
- Order Saga, Payment & Fulfillment
- Flash Sales, Hot Keys & Fair Admission
- Delivery Tracking & Event-Driven Updates

Applied AI Engineering

LLM apps, RAG, evaluation — secondary until core is strong. 77 sections · 471 study nodes.

E1 — Foundations & AI Tool Mastery4 sections

E1.1 — Applied AI Landscape & Engineering Roles6 nodes

Applied AI Engineering Landscape	TIER 1
— AI → ML → Deep Learning → Generative AI	
— Foundation, Language & Multimodal Models	
— AI Engineer vs ML Engineer vs Data Scientist vs Researcher	
— Frontier, Open-Weight, Small & Reasoning Models	
— Deterministic Software vs AI Decision	

E1.2 — AI Power-User & Coding-Agent Workflows7 nodes

AI Tool Mastery	TIER 1
— ChatGPT, Claude, Gemini & Research Tools	
— Compare Model Strengths, Limits & Outputs	
— Cursor, Copilot & Coding-Agent Workflows	
— Repository Research, Refactoring, Debugging & Tests	
— Delegate, Verify & Retain Human Ownership	
— Research Synthesis, Citations & Fact Checking	

E1.3 — Python Essentials for AI Engineers7 nodes

Python Essentials	TIER 1
— Collections	
— Functions, Iterators & Exceptions	
— Classes & Dataclasses	
— Type Hints	
— Modules & Package Management	
— Virtual Environments	

E1.4 — Python AI Backend & Data Stack7 nodes

Python AI Backend Engineering	TIER 1
— Pydantic Models & Validation	
— FastAPI & Typed REST APIs	
— Async/Await, SSE & WebSockets	
— Background Jobs & Queues	
— NumPy Basics	
— Pandas, Jupyter & Data Exploration	

E2 — ML, Deep Learning & LLM Internals8 sections

E2.1 — Machine Learning Foundations5 nodes

ML, Deep Learning & Generative AI	TIER 1
— Datasets, Features, Labels & Targets	
— Parameters vs Hyperparameters	
— Training vs Inference	

- Generalization, Distribution Shift & Baselines

E2.2 — Learning Paradigms & Core ML Problems

7 nodes

Learning Paradigms

TIER 1

- Supervised Learning
- Unsupervised Learning
- Self-Supervised & Semi-Supervised Learning
- Reinforcement Learning Concepts
- Regression & Classification
- Clustering, Ranking & Recommendation

E2.3 — Training, Generalization & Classical Evaluation

7 nodes

Training & Evaluation

TIER 1

- Train, Validation & Test Splits
- Loss Functions & Optimization
- Gradient Descent & Learning Rate
- Overfitting, Underfitting & Regularization
- Bias–Variance Trade-off
- Accuracy, Precision, Recall, F1 & ROC-AUC

E2.4 — Math for Applied AI — Just in Time

6 nodes

Math for Applied AI

TIER 2

- Vectors, Matrices, Tensors & Dot Products
- Probability, Bayes, Expectation & Sampling
- Statistics, Correlation & Confidence
- Derivatives, Gradients & Chain Rule
- SGD, Adam & Learning-Rate Schedules

E2.5 — Classical ML with scikit-learn

6 nodes

Classical ML Practice

TIER 2

- Linear & Logistic Regression
- Decision Trees, Random Forests & Gradient Boosting
- K-Means & PCA
- Preprocessing & scikit-learn Pipelines
- Cross-Validation & Hyperparameter Tuning

E2.6 — Neural Networks & PyTorch Fundamentals

6 nodes

Neural-Network Concepts

TIER 2

- Neurons, Weights, Biases & Activations
- Forward Propagation & Backpropagation
- PyTorch Tensors & Autograd
- nn.Module, DataLoaders & Training Loops
- CNN, RNN & LSTM Context

E2.7 — Transformers & Attention Internals

8 nodes

Transformer Architecture

TIER 1

- Tokenization
- Tokens, Vocabulary & Special Tokens
- Token Embeddings & Positional Information
- Query, Key, Value & Self-Attention
- Multi-Head & Causal Attention
- Residuals, LayerNorm, MLP & Transformer Blocks
- Encoder, Decoder & Encoder–Decoder Families

E2.8 — LLM Inference, Training Lifecycle & Limitations

7 nodes

Large Language Model Fundamentals

TIER 1

- Context Windows & Lost-in-the-Middle
- Logits, Temperature, Top-p & Sampling
- Prefill, Decode & KV Cache
- Pretraining, Instruction Tuning & Alignment
- Reasoning Models & Test-Time Compute
- Hallucinations, Bias & Reasoning Failures

E3 — LLM Application Engineering

5 sections

E3.1 — Provider APIs & Direct SDK Integration

6 nodes

LLM Provider APIs

TIER 1

- OpenAI API Patterns
- Anthropic API Patterns
- Gemini API Patterns
- Open-Model Inference APIs
- Thin Provider Abstraction without Framework Lock-in

E3.2 — Messages, Conversations & Streaming

6 nodes

Messages & Conversations

TIER 1

- System, Developer, User, Tool & Assistant Roles
- Conversation State & Stateless API Calls
- Streaming Responses
- Cancellation, Disconnects & Partial Output
- Usage Metadata & Token Accounting

E3.3 — Structured Outputs & Typed Extraction

6 nodes

Structured Outputs

TIER 1

- JSON Schema & Typed Responses
- Extraction, Classification & Transformation
- Validation, Parsing & Repair
- Refusals, Missing Fields & Partial Results
- Deterministic Post-Processing

E3.4 — Tool & Function Calling

6 nodes

Tool & Function Calling

TIER 1

- Tool Names, Descriptions & Argument Schemas
- Model → Tool → Result → Model Loop
- Parallel vs Sequential Tool Calls
- Argument Validation & Tool Error Recovery
- Authorization, Approval & Idempotency

E3.5 — Model Selection, Routing & API Reliability

6 nodes

Model Selection

TIER 1

- Quality, Latency & Cost Trade-offs
- Capability, Modality & Context Fit
- Provider Rate Limits
- Timeouts, Retries & Backoff
- Task Routing, Fallbacks & Provider Failover

E4 — Prompt & Context Engineering

5 sections

E4.1 — Prompt Fundamentals & Instruction Design

7 nodes

Prompt & Instruction Fundamentals

TIER 1

- Objective, Context, Constraints & Output Contract
- Zero-Shot Prompting
- Few-Shot & Example-Driven Prompting
- Structured Prompts & Delimiters
- Decomposition & Prompt Chaining
- Iterative Refinement & Critique

E4.2 — Context Engineering

6 nodes

Context Engineering

TIER 1

- Relevant Context at the Moment of Decision
- Context Selection, Ordering & Salience
- Compression, Summarization & Token Budgets
- Dynamic Context from Retrieval, Tools & State
- Context Pollution & Lost-in-the-Middle

E4.3 — Long Conversations & Memory Boundaries

6 nodes

Conversation Context

TIER 1

- Sliding-Window History
- Conversation Summarization
- Workflow State vs User Memory
- Provider Prompt / Prefix Caching
- Forgetting, Retention & User Control

E4.4 — Prompt Operations & Versioning

5 nodes

Prompt Operations

TIER 2

- Templates, Variables & Reusable Components
- Prompt Unit Tests & Regression Evals
- Environment-Specific Prompts
- Prompt/Model Version Tracing

E4.5 — Output Validation & Prompt-Injection Awareness

6 nodes

Output Validation

TIER 1

- Prompt-Injection Awareness
- Instructions vs Untrusted Content
- Typed Schemas & Allow-Listed Values
- Model Output Is Data, Not Authorization
- Safe Rendering, Escaping & Downstream Use

E5 — Embeddings & Semantic Search

4 sections

E5.1 — Embedding Models & Representation Spaces

5 nodes

Embeddings

TIER 1

- Text, Image & Multimodal Embeddings
- Dimensions, Normalization & Model Compatibility
- Domain, Language & Task Fit
- Embedding Versioning & Re-Indexing

E5.2 — Similarity & Nearest-Neighbor Search

6 nodes

Vector Similarity

TIER 1

- Cosine Similarity
- Dot Product vs Euclidean Distance
- Exact Nearest Neighbors
- Approximate Nearest Neighbors

- Recall, Latency & Memory Trade-offs

E5.3 — Vector Indexes & Storage

6 nodes

Vector Indexes

TIER 2

- HNSW
- IVF & Product Quantization Concepts
- PostgreSQL + pgvector
- Pinecone, Qdrant, Weaviate & Chroma
- Index Build, Updates, Deletes & Compaction

E5.4 — Semantic, Keyword & Hybrid Search

6 nodes

Semantic Search

TIER 1

- Keyword Search & BM25
- Dense + Sparse Hybrid Search
- Metadata & Access-Control Filtering
- Cross-Encoder Reranking
- Search Relevance Evaluation

E6 — Retrieval-Augmented Generation

8 sections

E6.1 — RAG Architecture & Knowledge Ingestion

6 nodes

RAG Ingestion

TIER 1

- Ingest → Index → Retrieve → Augment → Generate
- PDF, HTML, DOCX, Spreadsheet & OCR Parsing
- Cleaning, Normalization & Boilerplate Removal
- Metadata, Provenance & Source URLs
- Incremental Sync, Deletes & Reprocessing

E6.2 — Chunking & Document Structure

6 nodes

Chunking Strategies

TIER 1

- Fixed, Recursive & Structure-Aware Chunking
- Semantic Chunking
- Parent-Child & Small-to-Big Retrieval
- Overlap, Boundaries & Context Loss
- Evaluate Chunk Size by Task

E6.3 — Embedding & Indexing Pipelines

6 nodes

RAG Embedding & Indexing

TIER 1

- Indexing Pipeline
- Batching, Rate Limits & Retry
- Chunk, Document & ACL Index Schema
- Versioning & Zero-Downtime Re-Embedding
- Freshness, CDC & Event-Driven Indexing

E6.4 — Retrieval & Reranking

6 nodes

RAG Retrieval

TIER 1

- Top-K & Candidate Recall
- Metadata / Tenant / ACL Filtering
- Dense + Sparse Hybrid Retrieval
- Reranking & Score Fusion
- Diversity, Deduplication & MMR

E6.5 — Query Understanding & Retrieval Planning

6 nodes

RAG Query Processing

TIER 1

- Query Rewriting & Normalization
- Multi-Query Retrieval
- Query Decomposition
- HyDE Concepts
- Route by Corpus, Modality & Intent

E6.6 — Context Construction, Grounding & Citations

6 nodes

RAG Context Construction

TIER 1

- Grounding & Citations
- Context Ranking & Ordering
- Context Compression & Token Budget
- Claim-to-Source Citation Mapping
- Abstain on Insufficient Evidence

E6.7 — Advanced, Agentic & Multimodal RAG

6 nodes

Advanced RAG

TIER 2

- Agentic Retrieval
- Corrective & Adaptive RAG
- Graph RAG Concepts
- Hierarchical Retrieval
- Multimodal RAG

E6.8 — RAG Evaluation & Failure Modes

7 nodes

RAG Evaluation

TIER 1

- RAG Failure Modes
- Retrieval Precision, Recall & NDCG
- Context Relevance & Coverage
- Faithfulness & Answer Relevance
- Citation Correctness & Completeness
- Poisoning & Access-Control Tests

E7 — Agents, Workflows & MCP

10 sections

E7.1 — Agents vs Deterministic Workflows

6 nodes

Agent Fundamentals

TIER 1

- Agent vs Workflow
- Goal → Decide → Act → Observe → Complete
- When Not to Use an Agent
- Autonomy Spectrum & Risk
- Termination Conditions & Max Steps

E7.2 — Planning, Decomposition & Verification

6 nodes

Agent Planning

TIER 1

- Task Decomposition
- Dynamic Planning & Replanning
- Plan–Execute–Verify
- Reflection & Critique—Use Carefully
- Deterministic Verification & Assertions

E7.3 — Agent Workflow Patterns

7 nodes

Multi-Step Agent Workflows

TIER 1

- Sequential Workflows
- Parallel Fan-out / Fan-in
- Classification & Routing
- Orchestrator–Worker

- Evaluator-Optimizer
- Human-in-the-Loop

E7.4 — Agent State, Memory & Context

6 nodes

Agent State

TIER 1

- Working, Semantic & Episodic Memory
- Conversation Windows & Summaries
- Durable Checkpoints & Resumption
- Memory Extraction, Retrieval & Update
- Forgetting, Conflicts, Privacy & Retention

E7.5 — Agent Tool Use, Permissions & Approvals

6 nodes

Agent Tool Calling

TIER 1

- Tool Discovery & Selection
- Least-Privilege Tool Permissions
- Approval Gates for Consequential Actions
- Sandboxing & Isolation
- Idempotent Actions & Duplicate Prevention

E7.6 — Reliable Long-Running Agents

6 nodes

Agent Reliability

TIER 1

- Timeouts, Retries & Backoff
- Loop Detection & Step Budgets
- Partial Progress & Compensating Actions
- Checkpoint, Resume & Replay
- Escalation & Honest Failure

E7.7 — Multi-Agent Systems

6 nodes

Multi-Agent Architectures

TIER 2

- Supervisor & Specialist Agents
- Delegation & Handoffs
- Shared State & Communication
- Coordination Evals & Failure Modes
- One Capable Agent vs Multi-Agent

E7.8 — MCP Architecture & Primitives

6 nodes

Model Context Protocol Fundamentals

TIER 2

- Host, Client & Server
- MCP Tools
- MCP Resources
- MCP Prompts
- MCP vs Direct Function Calling

E7.9 — Building MCP Servers & Clients

6 nodes

Build MCP Integrations

TIER 2

- Server Implementation & Capability Discovery
- Client Connection & Invocation Lifecycle
- Errors, Timeouts & Cancellation
- Authentication & Authorization
- Remote MCP & Enterprise Integrations

E7.10 — Production MCP Security & Operations

6 nodes

Production MCP

TIER 2

- Trust Boundaries & Consent
- Tenant Isolation & Scoped Credentials
- Tool Tracing, Audit & Cost

- Schema Versioning & Compatibility
- Prompt Injection through Tools & Resources

E8 — Advanced Models & Multimodal AI

8 sections

E8.1 — Agent & RAG Frameworks—After Fundamentals

6 nodes

AI Application Frameworks

TIER 2

- Workflow Orchestration Frameworks
- LangGraph State, Nodes, Edges & Persistence
- LlamaIndex Ingestion & Retrieval
- Provider Agent SDKs
- What Frameworks Abstract—and Hide

E8.2 — Hugging Face & Open-Weight Ecosystem

6 nodes

Open-Weight AI

TIER 2

- Hugging Face Hub, Models & Datasets
- Transformers & Tokenizers Libraries
- Weights, Checkpoints & Safetensors
- Licenses, Commercial Use & Supply Chain
- Hosted Frontier vs Open-Weight Decision

E8.3 — Local Models, Formats & Quantization

6 nodes

Run Models Locally

TIER 2

- Ollama
- llama.cpp
- GGUF & Quantized Model Formats
- FP16/BF16, INT8 & INT4 Quantization
- Quality, Memory & Throughput Trade-offs

E8.4 — Model Serving & Inference Optimization

6 nodes

GPU Inference & Model Serving

TIER 2

- vLLM & Continuous Batching
- Prefill vs Decode
- KV Cache & Prefix Caching
- Speculative Decoding
- Tokens/sec, TTFT, Throughput & Tail Latency

E8.5 — Fine-Tuning & Model Customization

7 nodes

Fine-Tuning

TIER 2

- RAG vs Tools vs Fine-Tuning Decision
- SFT & Instruction Dataset Quality
- LoRA, QLoRA & PEFT
- Preference Data, RLHF & DPO Concepts
- Distillation & Synthetic Data
- Evaluate Behavior without Regressing Capability

E8.6 — Multimodal Document & Vision AI

6 nodes

Multimodal AI

TIER 2

- Vision–Language Models
- OCR, Layout & Document Understanding
- Screenshots, Charts & Diagram Reasoning
- Diffusion, Generation, Editing & Inpainting

E8.7 — Speech, Realtime & Voice Agents

6 nodes

Realtime Voice AI

TIER 2

- Speech-to-Text, Text-to-Speech & Speech-to-Speech
- Voice Activity Detection & Turn Taking
- Interruptions & Barge-In
- Realtime Tool Calling
- End-to-End Latency & Streaming Audio

E8.8 — AI Hardware & Distributed Inference

7 nodes

AI Infrastructure

TIER 3

- CPU vs GPU, VRAM & Memory Bandwidth
- CUDA Kernels & Parallelism Concepts
- Data Parallelism
- Tensor Parallelism
- Pipeline Parallelism
- Distributed Inference & Model Placement

E9 — AI Evaluation & Testing

5 sections

E9.1 — Evaluation Strategy & Golden Datasets

6 nodes

AI Evaluation Foundations

TIER 1

- Why Probabilistic Software Needs Evals
- Curated, Production & Synthetic Test Cases
- Rubrics, Failure Taxonomy & Dataset Slices
- Dataset Versioning & Leakage Prevention
- Evaluate before Prompt/Model/RAG Changes

E9.2 — LLM Output Evaluation

6 nodes

Offline LLM Evaluations

TIER 1

- Answer Correctness & Task-Specific Metrics
- Groundedness & Hallucination Measurement
- Exact Match vs Semantic Similarity
- LLM-as-Judge, Rubrics & Calibration
- Human Evaluation & Inter-Rater Agreement

E9.3 — Retrieval & RAG Evaluation

6 nodes

Retrieval Evaluation

TIER 1

- Recall@K, Precision@K, MRR & NDCG
- Context Relevance & Faithfulness
- Citation Accuracy & Coverage
- Component vs End-to-End RAG Evals
- Chunking/Retrieval/Reranking Ablations

E9.4 — Agent & Tool-Use Evaluation

6 nodes

Agent Evaluation

TIER 1

- Tool Selection & Argument Correctness
- Task Success, Partial Credit & Side Effects
- Steps, Latency, Tokens & Cost
- Trajectory vs Final-Outcome Evaluation

- Permission, Injection & Destructive-Action Evals

E9.5 — Regression, Online Experiments & Feedback

6 nodes

AI Regression Testing

TIER 1

- Prompt, Model, Retrieval & Agent Regression
- A/B Testing & Guardrail Metrics
- Explicit/Implicit User Feedback
- Shadow, Canary & Progressive Rollout
- Production Samples → Evals → Improvement Loop

E10 — Production AI & Product Engineering

7 sections

E10.1 — Production AI Reference Architecture

6 nodes

Production AI Architecture

TIER 1

- AI Gateway & Orchestrator
- Models, Retrieval, Tools & Memory Boundaries
- Authentication, Tenant Context & Entitlements
- Version Prompts, Models, Indexes & Tools
- Quality, Latency, Reliability, Security & Cost

E10.2 — Streaming, Async Jobs & Backend Infrastructure

6 nodes

Streaming AI Responses

TIER 1

- Async Processing
- Queues & Workers
- SSE vs WebSocket vs Polling
- Cancellation, Backpressure & Disconnects
- Long-Running Job State & Resume

E10.3 — AI Performance, Caching & Context Efficiency

6 nodes

AI Performance & Caching

TIER 1

- Prompt / Prefix Caching
- Semantic Caching & Correctness Risk
- Token & Context Budgets
- Parallel Retrieval & Tool Calls
- Batching & Connection Pooling

E10.4 — Reliability, Limits & Graceful Degradation

7 nodes

Production AI Reliability

TIER 1

- Timeouts & Deadline Budgets
- Safe Retries & Backoff
- Rate Limiting & Admission Control
- Fallback Models & Deterministic Fallbacks
- Circuit Breakers & Provider Isolation
- Graceful Degradation & Human Escalation

E10.5 — Model Routing & Cost Engineering

7 nodes

Model Routing

TIER 1

- Cost per Request, User & Successful Task
- Small/Fast vs Frontier/Reasoning Models
- Rules, Classifiers & Dynamic Routing
- Batch Processing & Deferred Work
- Context Reduction without Quality Loss

E10.6 — AI Observability & Quality Monitoring

6 nodes

AI Observability

TIER 1

- Trace Model, Retrieval, Tool & Agent Steps
- Prompt, Model, Index & Tool Version Tracking
- Latency, Tokens, Cost & Error Metrics
- Quality, Groundedness & Task-Success Signals
- Feedback, Drift & Regression Detection

E10.7 — AI Product Engineering & Decision Framework

7 nodes

AI Product Engineering

TIER 1

- AI-Suitable vs Deterministic Problems
- UX for Uncertainty, Sources & Editable Outputs
- Approval, Retry & Human Escalation
- Task Completion, Acceptance, Retention & Cost/Task
- Build vs Buy & Provider Lock-In
- RAG vs Tools vs Agents vs Fine-Tuning vs Open Models

E11 — AI Security & Responsible AI

5 sections

E11.1 — Prompt Injection & Untrusted Content

6 nodes

Prompt Injection

TIER 1

- Indirect Prompt Injection
- Jailbreaks & Instruction Conflicts
- Separate Trusted Instructions from Untrusted Data
- Malicious Tool Output & Cross-Domain Injection
- Defense in Depth—not Prompt-Only Defenses

E11.2 — Agent Security, Agency & Tool Abuse

6 nodes

Agent & Tool Security

TIER 1

- Excessive Agency
- Least-Privilege Permission Boundaries
- The Model Is Never the Authorization Layer
- Human Approval for Consequential Actions
- Sandboxing, Idempotency & Audit Logs

E11.3 — Data, Tenant & RAG Security

9 nodes

AI Data Security

TIER 1

- Sensitive Data & PII Handling
- Production PII Detection & Redaction
- Provider and Tool Secrets
- Application Access Control
- Tenant-Isolated Retrieval & Memory
- Poisoned Documents & Knowledge Bases
- Access-Controlled Retrieval before Generation
- Retention, Encryption & Provider Data Policies

E11.4 — Input/Output Safety & Responsible AI

6 nodes

AI Input & Output Safety

TIER 1

- Schema Validation, Allowlists & Safe Rendering
- Harmful Content & Moderation Boundaries
- Bias, Fairness & Dataset Representation
- Transparency, Sources & User Disclosure

E11.5 — Adversarial Testing & AI Threat Modeling

6 nodes

Adversarial AI Testing

TIER 2

- Assets, Actors, Trust Boundaries & Abuse Cases
- Injection/Jailbreak Red-Team Dataset
- Tool Abuse & Exfiltration Tests
- Security Regression Evals
- AI Security Incident Response

E12 — Applied AI Project Ladder

8 sections

E12.1 — Project: Intelligent Document Analyzer

5 nodes

Document-Analysis Platform

TIER 1

- PDF, DOCX, Image, OCR & Layout
- Typed Extraction & Validation
- Questions, Citations & Suggested Actions
- Golden Evals, PII & Malicious Documents

E12.2 — Project: Production Enterprise RAG

5 nodes

Enterprise RAG & Search System

TIER 1

- 500+ Documents & Incremental Ingestion
- Hybrid Retrieval, Reranking & Citations
- ACL-Aware Multi-Tenant Retrieval
- RAG Evals, Tracing & Hallucination Handling

E12.3 — Project: AI Customer Support System

5 nodes

AI Customer-Support System

TIER 1

- Intent, Routing & Knowledge Retrieval
- CRM Tools & Ticket Actions
- Confidence, Approval & Human Escalation
- Resolution Quality, Latency & Cost/Case

E12.4 — Project: Repository-Aware Coding Assistant

5 nodes

Coding Assistant

TIER 1

- Repository Indexing & Context Selection
- Search, Edit, Test & Sandboxed Shell Tools
- Planning, Diff Review & Human Approval
- Patch Correctness, Test & Security Evals

E12.5 — Project: AI Research Assistant

5 nodes

AI Research Assistant

TIER 1

- Search, Fetch & Source Quality
- Multi-Source Synthesis & Contradictions
- Claim-Level Citations & Verification
- Long-Running Workflow, Checkpoints & Cost

E12.6 — Project: Tool-Using Productivity Agent

5 nodes

AI Agent with External Tools

TIER 1

- Calendar, Email, Tasks, Web & Weather Tools
- State, Memory & Resumable Workflows
- Scoped Credentials, Approval & Idempotency

- Task-Success, Tool & Safety Evals

E12.7 — Project: Recommendation & Personalization

5 nodes

Recommendation & Personalization System	TIER 2
— Candidate Generation & Ranking	
— Embeddings & Online Feedback	
— Cold Start, Exploration & Diversity	
— Bias, Business Metrics & A/B Tests	

E12.8 — Capstone: Multi-Tenant AI Executive Assistant

6 nodes

Multi-Tenant Production AI Platform	TIER 2
— Web, Mobile, Document & Voice Inputs	
— Model Router + RAG + Agent + MCP	
— Tenant Memory, Tools & Human Approval	
— Evals, Injection Defense & Access Control	
— Tracing, Reliability, Caching & Cost Budgets	